

Find the greatest common factor of 16 and 40.

GCF: _____

SET 21 • 6.NS.B.4

BACKWARD 1

Factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factors of 18.

Factors: _____

True or false?

7 is a factor of 42.

A. True

B. False

Which number is NOT a factor of 36?

- A.** 4
- B.** 8
- C.** 9
- D.** 12

True or false?

5 is a factor of 32.

A. True

B. False

List all the factors of 24.

Factors: _____

SET 21 • 6.NS.B.4

BACKWARD 2

Prime and composite numbers • Grade 4 • Operations & Algebraic Thinking •
4.OA.B.4

True or false?

13 is a prime number.

A. True

B. False

Which number is composite?

- A.** 2
- B.** 11
- C.** 17
- D.** 21

True or false?

1 is a prime number.

A. True

B. False

Is 29 prime or composite?

Answer: _____

Which number is prime?

A. 27

B. 31

C. 33

D. 39

SET 21 • 6.NS.B.4

BACKWARD 3

Division facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Divide.

$$42 \div 7$$

Quotient: _____

Divide.

$$36 \div 4$$

Quotient: _____

True or false?

$$45 \div 9 = 5$$

A. True

B. False

$$56 \div 8 = ?$$

A. 6

B. 7

C. 8

D. 9

Divide.

$$63 \div 9$$

Quotient: _____

SET 21 • 6.NS.B.4

FORWARD 1

Factor linear expressions using a common factor • Grade 7 • Expressions & Equations • 7.EE.A.1

Factor using the greatest common factor.

$$18x + 24$$

Factored form: _____

Factor using the greatest common factor.

$$20a - 35$$

Factored form: _____

Which shows $16m + 40$ factored using the greatest common factor?

- A.** $8(2m + 5)$
- B.** $2(8m + 20)$
- C.** $8(2m + 40)$
- D.** $4(4m + 10)$

Factor using the greatest common factor.

$$27x + 45y$$

Factored form: _____

True or false?

$$14n - 21 = 7(2n - 3)$$

A. True

B. False

SET 21 • 6.NS.B.4

FORWARD 2

Solve equations using the distributive property • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$12x + 18 = 6(x + 5)$$

x = _____

Solve for x.

$$4(3x - 2) = 8(x + 1)$$

x = _____

Solve for x.

$$15x - 10 = 5(2x + 3)$$

A. -1

B. 1

C. 5

D. 25

Solve for x.

$$9(x + 2) = 6(x + 5)$$

x = _____

Solve for x.

$$10x + 25 = 5(x + 10)$$

x = _____

Find the least common multiple of two whole numbers

MAIN	Grade 6 main question	6.NS.B.4
BACKWARD 1	Multiples	4.OA.B.4
BACKWARD 2	Multiplication facts	3.OA.C.7
FORWARD 1	Add and subtract rational numbers with unlike denominators	7.NS.A.1.d
FORWARD 2	Equations with fractions: clearing denominators	8.EE.C.7.b

SET 22 • 6.NS.B.4

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.4

Find the least common multiple of 6 and 8.

LCM: _____

Find the least common multiple of 4 and 10.

LCM: _____

What is the least common multiple of 9 and 12?

- A.** 3
- B.** 36
- C.** 72
- D.** 108

Find the least common multiple of 5 and 7.

LCM: _____

Find the least common multiple of 6 and 15.

LCM: _____

SET 22 • 6.NS.B.4

BACKWARD 1

Multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first five multiples of 6.

Multiples: _____

True or false?

36 is a multiple of 8.

A. True

B. False

Which number is a multiple of both 4 and 6?

- A.** 8
- B.** 12
- C.** 18
- D.** 20

True or false?

45 is a multiple of 9.

A. True

B. False

What is the 4th multiple of 7?

Multiple: _____

SET 22 • 6.NS.B.4

BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$6 \times 8$$

Product: _____

Multiply.

$$9 \times 4$$

Product: _____

True or false?

$$7 \times 5 = 35$$

A. True

B. False

$$3 \times 12 = ?$$

A. 15

B. 33

C. 36

D. 39

Multiply.

$$5 \times 6$$

Product: _____

SET 22 • 6.NS.B.4

FORWARD 1

Add and subtract rational numbers with unlike denominators • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-\frac{5}{6} + \frac{3}{8}$$

Find the value.

$$\frac{3}{4} - \left(-\frac{1}{6}\right)$$

$$-\frac{2}{3} - \frac{1}{4} = ?$$

A. $-\frac{11}{12}$

B. $-\frac{3}{7}$

C. $-\frac{1}{12}$

D. $\frac{5}{12}$

Find the value.

$$-1\frac{1}{2} + \frac{5}{6}$$

True or false?

$$\frac{3}{10} + \left(-\frac{2}{15}\right) = \frac{1}{6}$$

A. True

B. False

SET 22 • 6.NS.B.4

FORWARD 2

Equations with fractions: clearing denominators • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$\frac{x}{6} + \frac{x}{8} = 7$$

x = _____

Solve for x.

$$\frac{x}{4} - 1 = \frac{x}{6} + 2$$

x = _____

Solve for x.

$$\frac{2x}{3} + \frac{x}{2} = 14$$

A. 7

B. 12

C. $\frac{84}{5}$

D. 42

Solve for x.

$$\frac{x}{5} + 3 = \frac{x}{2}$$

x = _____

Solve for x.

$$\frac{(x+1)}{4} = \frac{(x-2)}{3}$$

x = _____

Use the distributive property to write a sum as the GCF times a sum

MAIN	Grade 6 main question	6.NS.B.4
BACKWARD 1	Common factors	4.OA.B.4
BACKWARD 2	Distributive property with whole numbers	3.OA.B.5
BACKWARD 3	Parentheses and order of operations	5.OA.A.1
FORWARD 1	Factor and expand linear expressions	7.EE.A.1
FORWARD 2	One, none, or infinitely many solutions	8.EE.C.7.a

SET 23 • 6.NS.B.4

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.4

Use the greatest common factor to write $45 + 20$ as a product of the GCF and a sum of two whole numbers.

Expression: _____

Which expression is equal to $42 + 56$ and uses the greatest common factor?

A. $2(21 + 28)$

B. $14(3 + 4)$

C. $14(3 + 8)$

D. $7(6 + 8)$

Use the greatest common factor to write $27 + 45$ as a product.

Expression: _____

True or false?

$$24 + 60 = 12(2 + 5)$$

A. True

B. False

Complete the equation.

$$30 + 75 = 15(\underline{\quad} + \underline{\quad})$$

First blank: _____

Second blank: _____

SET 23 • 6.NS.B.4

BACKWARD 1

Common factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

True or false?

5 is a factor of both 45 and 20.

A. True

B. False

List the factors of 12 and the factors of 18.

Which factors do they share?

Shared factors: _____

Which number is a factor of both 27 and 45?

- A.** 5
- B.** 6
- C.** 9
- D.** 15

List all the factors of 42.

Factors: _____

True or false?

6 is a factor of 56.

A. True

B. False

SET 23 • 6.NS.B.4

BACKWARD 2

Distributive property with whole numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$9 \times 6 = 9 \times 4 + 9 \times 2$$

A. True

B. False

True or false?

$$4 \times (9 + 2) = 4 \times 9 + 2$$

A. True

B. False

Which is equal to 6×12 ?

A. $6 \times 10 + 2$

B. $6 \times 10 \times 2$

C. $6 \times 10 + 6 \times 2$

D. $6 + 10 \times 2$

Fill in the blank.

$$5 \times 13 = 5 \times 10 + 5 \times \underline{\quad}$$

Blank: _____

Fill in the blank.

$$3 \times (4 + 5) = 3 \times 4 + 3 \times \underline{\hspace{1cm}}$$

Blank: _____

SET 23 • 6.NS.B.4

BACKWARD 3

Parentheses and order of operations • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$5 \times (9 + 4)$$

Value: _____

Evaluate.

$$14 \times (3 + 4)$$

Value: _____

True or false?

$$9 \times (3 + 5) = 32$$

A. True

B. False

$$12 \times (2 + 5) = ?$$

A. 17

B. 29

C. 60

D. 84

Evaluate.

$$(20 - 5) \times 3$$

Value: _____

SET 23 • 6.NS.B.4

FORWARD 1

Factor and expand linear expressions • Grade 7 • Expressions & Equations •
7.EE.A.1

Factor.

$$12x - 18$$

Factored form: _____

Expand.

$$-3(2x - 5)$$

Expanded form: _____

Which expression is equivalent to $8y + 20$?

A. $4(2y + 20)$

B. $8(y + 20)$

C. $2(4y + 20)$

D. $4(2y + 5)$

Factor.

$$15a + 10b - 5$$

Factored form: _____

True or false?

$$\frac{1}{2}(6x - 4) = 3x - 2$$

A. True

B. False

SET 23 • 6.NS.B.4

FORWARD 2

One, none, or infinitely many solutions • Grade 8 • Expressions & Equations •
8.EE.C.7.a

How many solutions does the equation have?

$$4(x + 3) = 4x + 12$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$2(3x - 1) = 6x + 5$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$3(x + 2) = 2x + 9$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

What number goes in the box so that the equation has infinitely many solutions?

$$5(x - 2) = 5x - \square$$

$$\square = \underline{\hspace{10cm}}$$

Which equation has no solution?

A. $2(x + 4) = 2x + 5$

B. $2(x + 4) = x + 8$

C. $2(x + 4) = 2x + 8$

D. $2(x + 4) = 3x$

Use positive and negative numbers to represent quantities in context

MAIN	Grade 6 main question	6.NS.C.5
BACKWARD 1	Whole numbers as distances from 0	2.MD.B.6
BACKWARD 2	Increase and decrease situations	2.OA.A.1
FORWARD 1	Adding integers as moving on a number line	7.NS.A.1.b
FORWARD 2	The sign of the rate of change in context	8.F.B.4

SET 24 • 6.NS.C.5

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.5

A diver is 30 feet below sea level.

Write an integer to represent the diver's position.

Integer: _____

A company lost \$2,500 this month.

Write an integer to represent this change.

Integer: _____

A bank account shows a withdrawal of \$45.
Which number represents this change?

- A.** -45
- B.** -4.5
- C.** 0
- D.** 45

The temperature rose 12 degrees.

Write an integer to represent this change.

Integer: _____

A hawk flies 120 feet above sea level. A submarine is 120 feet below sea level.

Write an integer for each position.

Hawk: _____

Submarine: _____

SET 24 • 6.NS.C.5

BACKWARD 1

Whole numbers as distances from 0 • Grade 2 • Measurement & Data •
2.MD.B.6

What number does point A represent?



Number: _____

How many units from 0 is the number 6?

Units: _____

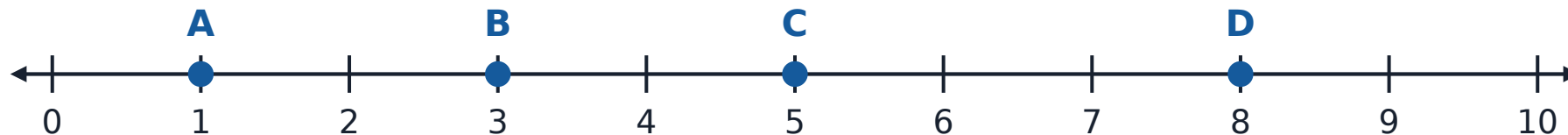
True or false?

On a number line, 9 is farther from 0 than 4 is.

A. True

B. False

Which point is 3 units from 0?



A. Point A

B. Point B

C. Point C

D. Point D

What number does point B represent?



Number: _____

SET 24 • 6.NS.C.5

BACKWARD 2

Increase and decrease situations • Grade 2 • Operations & Algebraic Thinking •
2.OA.A.1

Kim had \$20. She spent \$8.

How much money does she have now?

Money: _____

A plant was 12 cm tall. It grew 5 cm.

How tall is the plant now?

Height: _____

Which situation describes a decrease?

- A.** A plant grows 4 inches.
- B.** Tom earns \$10.
- C.** A team gains 5 yards.
- D.** A balloon loses 6 feet of height.

True or false?

"The temperature fell 7 degrees" describes a decrease.

A. True

B. False

Which word describes an increase in money?

- A.** Deposit
- B.** Loss
- C.** Drop
- D.** Withdrawal

SET 24 • 6.NS.C.5

FORWARD 1

Adding integers as moving on a number line • Grade 7 • The Number System •
7.NS.A.1.b

The temperature is -4°F . It rises 9°F .

What is the new temperature?

Temperature: _____

A submarine is at -120 feet. It rises 45 feet.
What is its new position?

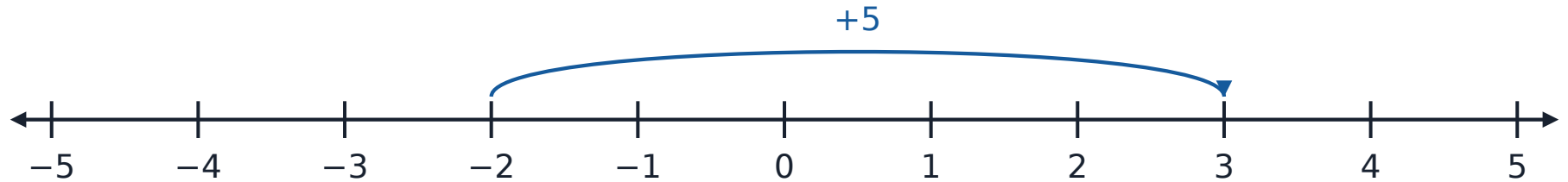
Position: _____

On a number line, where is $6 + (-4)$ compared to 6?

- A.** 4 units to the right of 6
- B.** 4 units to the left of 6
- C.** At 0
- D.** 6 units to the left of 4

The number line shows $-2 + 5$.

What is the sum?



Sum: _____

True or false?

A \$30 debt plus a \$30 deposit makes a balance of \$0.

A. True

B. False

SET 24 • 6.NS.C.5

FORWARD 2

The sign of the rate of change in context • Grade 8 • Functions • 8.F.B.4

A tank holds 60 gallons. It drains 3 gallons per minute.
Write a function for the amount A after t minutes.

$A =$ _____

An airplane at 12,000 feet descends 800 feet per minute.

What is the rate of change of its altitude?

- A.** $-12,000$ feet per minute **B.** -800 feet per minute
C. 800 feet per minute **D.** $12,000$ feet per minute

The table shows a linear function.

What is the rate of change in degrees per hour?

Hours	0	2	4
Temperature (°F)	5	1	-3

Rate of change: _____

A diver starts at -10 feet and descends 4 feet per second.

Write a function for the diver's position d after t seconds.

$d =$ _____

True or false?

A line through $(0, 8)$ and $(4, 0)$ has a rate of change of -2 .

A. True

B. False

Explain the meaning of 0 in a real-world situation

MAIN	Grade 6 main question	6.NS.C.5
BACKWARD 1	Zero as the starting point on a number line	2.MD.B.6
BACKWARD 2	A change of zero means no change	1.OA.D.8
FORWARD 1	Opposite quantities combine to make 0	7.NS.A.1.a
FORWARD 2	Interpret the initial value (the value when $x = 0$)	8.F.B.4

SET 25 • 6.NS.C.5

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.5

On a Celsius thermometer, what does 0°C represent?

- A.** No temperature at all
- B.** The temperature at which water freezes
- C.** The hottest possible temperature
- D.** The temperature at which water boils

When numbers describe elevation, what does 0 represent?

- A.** Sea level
- B.** No elevation at all
- C.** The top of a mountain
- D.** The ocean floor

A bank account has a balance of \$0.

What does this mean?

- A.** The account has no money and owes nothing.
- B.** The account owes money.
- C.** The account has money in it.
- D.** The account earns interest.

A football team's change in yards on a play is 0.

What does 0 mean?

- A.** The team scored.
- B.** The team gained 10 yards.
- C.** The team neither gained nor lost yards.
- D.** The team lost all its yards.

A scale shows that your weight changed by 0 pounds since last month.

What does 0 mean in this situation?

Meaning: _____

SET 25 • 6.NS.C.5

BACKWARD 1

Zero as the starting point on a number line • Grade 2 • Measurement & Data •
2.MD.B.6

True or false?

On a number line, 0 is the starting point for measuring distance.

- A.** True
- B.** False

What number is at point A?



A = _____

How far is 0 from itself on a number line?

- A.** 0 units
- B.** 1 unit
- C.** 2 units
- D.** 10 units

How far is 5 from 0 on a number line?

Units: _____

True or false?

A ruler starts measuring at 1.

A. True

B. False

SET 25 • 6.NS.C.5

BACKWARD 2

A change of zero means no change • Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8

What number goes in the box?

$$8 + \square = 8$$

$$\square = \underline{\hspace{10cm}}$$

What number goes in the box?

$$15 - \square = 15$$

$$\square = \underline{\hspace{10cm}}$$

True or false?

A plant was 12 cm tall last week and is 12 cm tall now. It grew 0 cm.

A. True

B. False

A score went from 45 to 45.

What was the change?

- A.** 0
- B.** 1
- C.** 45
- D.** 90

What number goes in the box?

$$\square + 23 = 23$$

$$\square = \underline{\hspace{10cm}}$$

SET 25 • 6.NS.C.5

FORWARD 1

Opposite quantities combine to make 0 • Grade 7 • The Number System •
7.NS.A.1.a

Maya earns \$25 and then spends \$25.

What is the total change in her money?

Total change: _____

True or false?

$$-8 + 8 = 0$$

A. True

B. False

Which sum is equal to 0?

A. $-3.5 + 3.5$

B. $-3.5 + 0$

C. $-3.5 + (-3.5)$

D. $3.5 + 3.5$

A balloon rises 40 feet and then falls 40 feet.
What is its total change in height?

Total change: _____

What number added to 14.2 gives 0?

Number: _____

SET 25 • 6.NS.C.5

FORWARD 2

Interpret the initial value (the value when $x = 0$) • Grade 8 • Functions •
8.F.B.4

The water in a tank is $W = 60 - 3t$, where t is minutes.

What does 60 represent?

Meaning: _____

A hiker's elevation is $E = -40 + 15t$, where t is hours.

What does -40 represent?

- A.** The hiker starts 40 m below sea level.
- B.** The hiker ends 40 m below sea level.
- C.** The hiker climbs 40 m each hour.
- D.** The hike lasts 40 hours.

A phone's battery charge is $B = 100 - 8h$ after h hours.

What is B when $h = 0$, and what does it mean?

Value: _____

Meaning: _____

A savings account has $y = 25x + 150$ dollars after x weeks.

What does 150 represent?

- A.** The starting amount
- B.** The amount saved each week
- C.** The number of weeks
- D.** The amount after 25 weeks

True or false?

For the function $y = 4x$, the output is 0 when the input is 0.

A. True

B. False

Find opposites; the opposite of the opposite of a number

MAIN	Grade 6 main question	6.NS.C.6.a
BACKWARD 1	Distance from 0 on a number line	2.MD.B.6
BACKWARD 2	Positive and negative numbers in context	6.NS.C.5
FORWARD 1	Subtract by adding the opposite	7.NS.A.1.c
FORWARD 2	Solve equations by adding opposites to both sides	8.EE.C.7.b

SET 26 • 6.NS.C.6.a

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of -7 ?

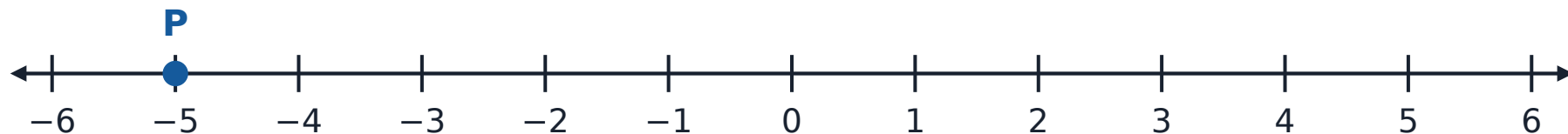
Opposite: _____

Simplify.

$$-(-12)$$

Value: _____

Which number is the opposite of the number at point P?



A. -5

B. 0

C. $\frac{1}{5}$

D. 5

True or false?

The opposite of 0 is 0.

A. True

B. False

What is the opposite of the opposite of 9?

Answer: _____

SET 26 • 6.NS.C.6.a

BACKWARD 1

Distance from 0 on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

How far is point A from 0?



Units: _____

How many units from 0 is the number 8?

Units: _____

Which number is 2 units from 0 on this number line?



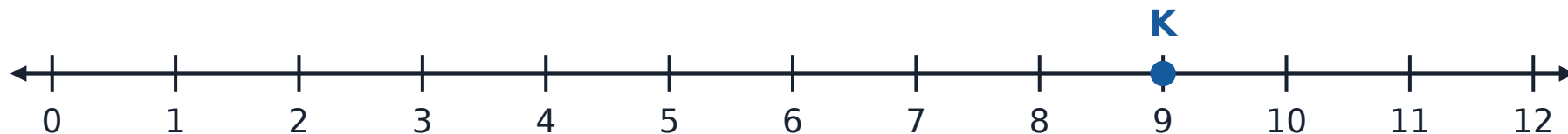
A. 0

B. 2

C. 4

D. 20

What number does point K represent?



Number: _____

True or false?

The distance from 0 to 7 is 7 units.

A. True

B. False

SET 26 • 6.NS.C.6.a

BACKWARD 2

Positive and negative numbers in context • Grade 6 • The Number System •
6.NS.C.5

True or false?

–5 can represent 5 degrees below zero.

A. True

B. False

Write an integer for a gain of 12 yards.

Integer: _____

Which integer represents 20 feet below sea level?

A. -20

B. -2

C. 0

D. 20

Write an integer for a loss of \$7.

Integer: _____

True or false?

A deposit of \$15 is written as -15 .

A. True

B. False

SET 26 • 6.NS.C.6.a

FORWARD 1

Subtract by adding the opposite • Grade 7 • The Number System • 7.NS.A.1.c

Rewrite $5 - 8$ as an addition expression.

Expression: _____

Find the value.

$$-3 - (-7)$$

Value: _____

True or false?

$$6 - (-2) = 6 + 2$$

A. True

B. False

Which expression is equal to $-4 - 9$?

A. $4 + 9$

B. $4 - (-9)$

C. $-4 + 9$

D. $-4 + (-9)$

Find the value.

$$2.5 - 6$$

Value: _____

SET 26 • 6.NS.C.6.a

FORWARD 2

Solve equations by adding opposites to both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$5x - 7 = 2x + 8$$

x = _____

Solve for x.

$$-3x + 4 = 3x - 8$$

x = _____

Solve for x.

$$6 - 2x = 4x - 12$$

A. -3

B. 1

C. 3

D. 9

Solve for x.

$$7x + 3 = -2x + 21$$

x = _____

Solve for x.

$$4(x - 1) = 2x + 10$$

x = _____

Use signs of coordinates to identify quadrants

MAIN	Grade 6 main question	6.NS.C.6.b
BACKWARD 1	Axes, origin, and ordered pairs	5.G.A.1
BACKWARD 2	Negative numbers are on the opposite side of 0	6.NS.C.6.a
BACKWARD 3	Reading points in the first quadrant	5.G.A.2
FORWARD 1	Signs of products of rational numbers	7.NS.A.2.a
FORWARD 2	Transformations and the quadrant of the image	8.G.A.3

SET 27 • 6.NS.C.6.b

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.b

In which quadrant is the point $(-4, 7)$?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

In which quadrant is the point $(3, -8)$?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

In which quadrant is the point $(-2.5, -6)$?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

A point has a negative x-coordinate and a positive y-coordinate.

In which quadrant is it?

Quadrant: _____

Which point is in Quadrant III?

A. $(-5, -1)$

B. $(5, 1)$

C. $(5, -1)$

D. $(-5, 1)$

SET 27 • 6.NS.C.6.b

BACKWARD 1

Axes, origin, and ordered pairs • Grade 5 • Geometry • 5.G.A.1

True or false?

The origin is the point $(0, 0)$.

A. True

B. False

In the ordered pair (5, 2), which number tells how far to move along the x-axis?

A. 2

B. 3

C. 5

D. 7

What is the name of the point where the x-axis and the y-axis meet?

Name: _____

True or false?

In the ordered pair (3, 8), the y-coordinate is 3.

A. True

B. False

Which axis is horizontal?

A. The x-axis

B. The y-axis

SET 27 • 6.NS.C.6.b

BACKWARD 2

Negative numbers are on the opposite side of 0 • Grade 6 • The Number System • 6.NS.C.6.a

True or false?

On a horizontal number line, -4 is to the left of 0 .

- A. True
- B. False

On a vertical number line, where is -3 ?

- A.** Above 0
- B.** Above 3
- C.** Below 0
- D.** At 0

True or false?

On a vertical number line, 5 is below 0.

A. True

B. False

On a horizontal number line, is 2.5 to the left or the right of 0?

Answer: _____

Which number is to the left of 0 on a horizontal number line?

A. -6

B. 0

C. $\frac{1}{2}$

D. 6

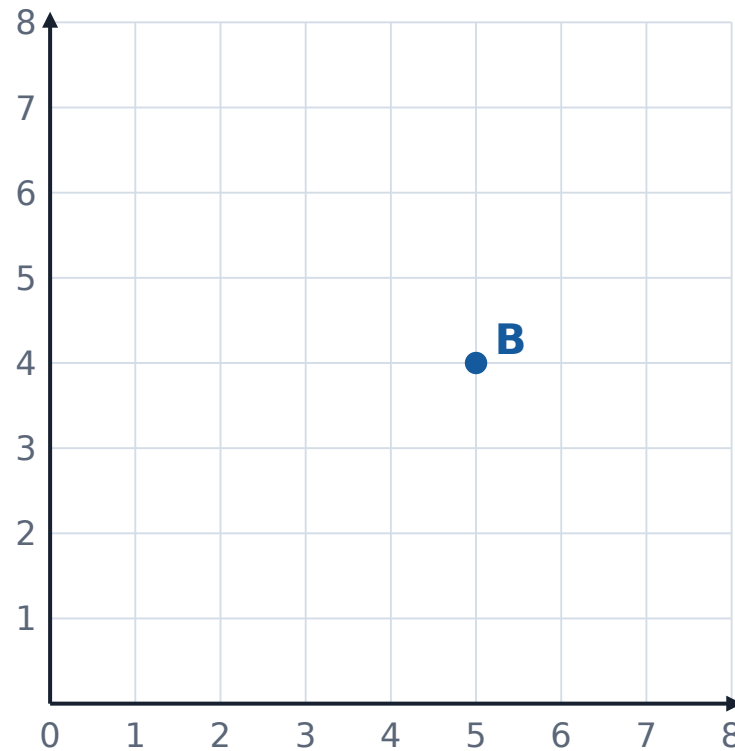
SET 27 • 6.NS.C.6.b

BACKWARD 3

Reading points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

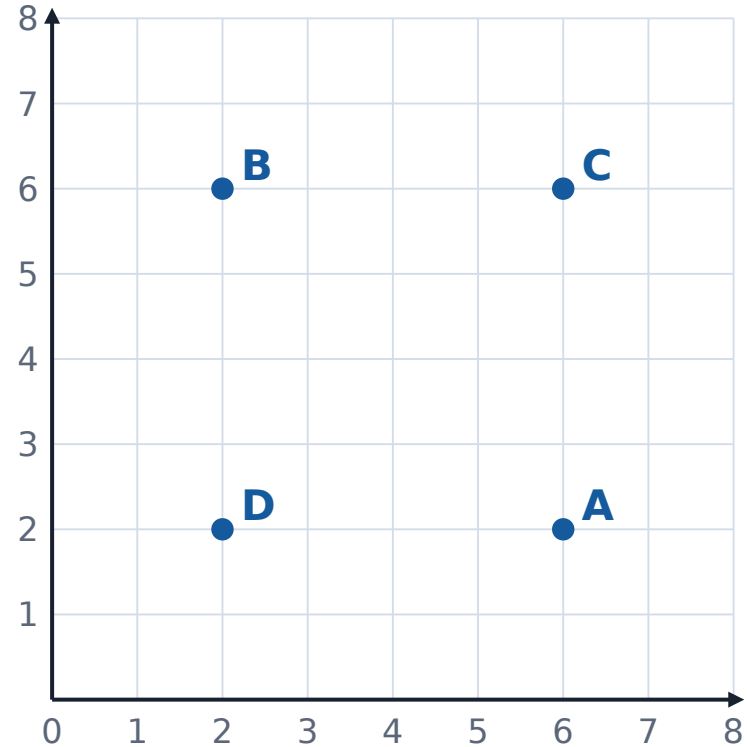
What are the coordinates of point B?

B = _____



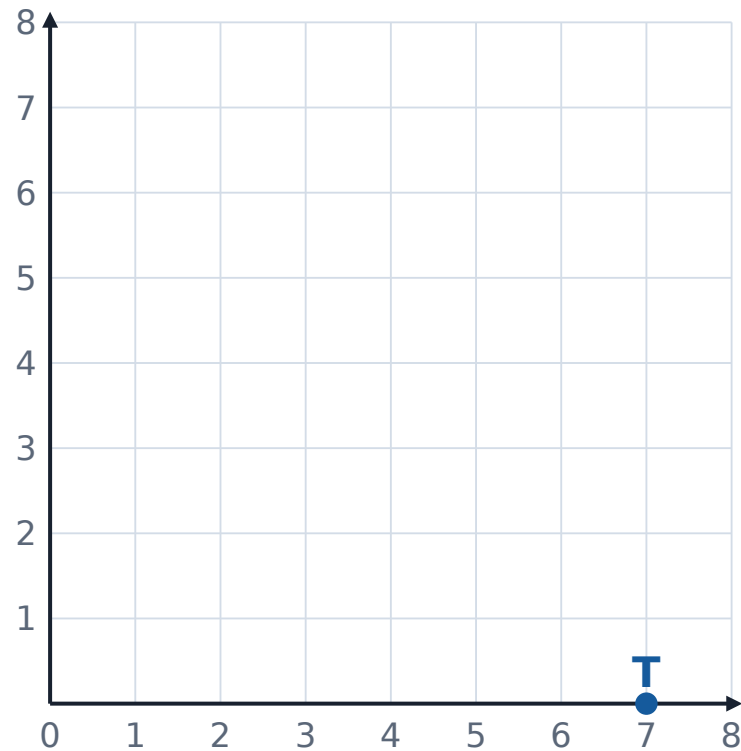
Which point is located at (2, 6)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D



What are the coordinates of point T?

T = _____

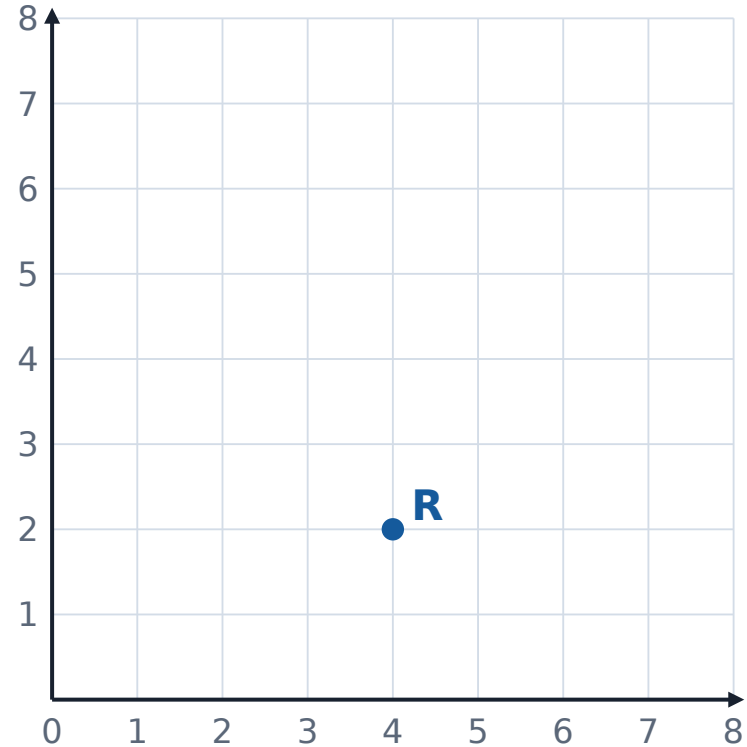


True or false?

Point R is located at (2, 4).

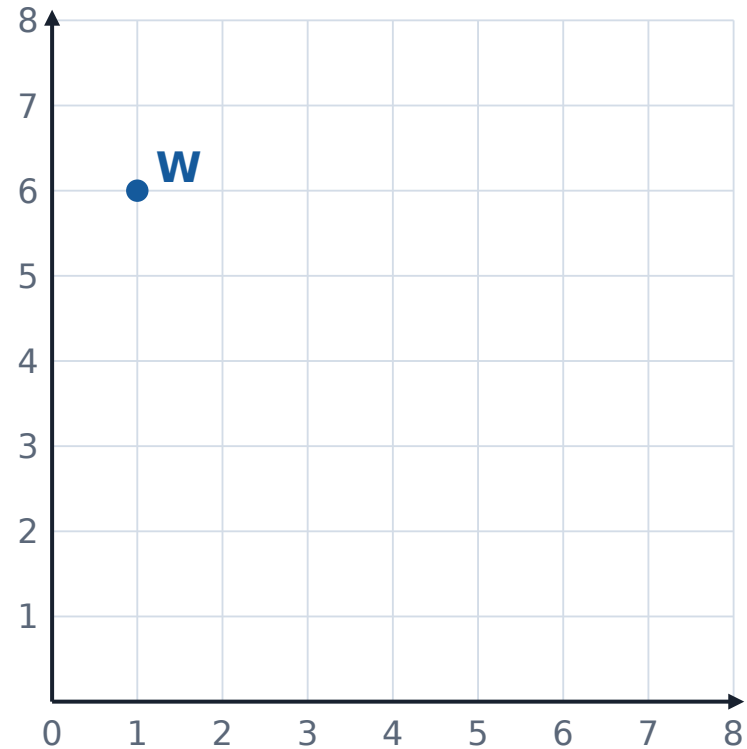
A. True

B. False



What are the coordinates of point W?

W = _____



SET 27 • 6.NS.C.6.b

FORWARD 1

Signs of products of rational numbers • Grade 7 • The Number System •
7.NS.A.2.a

The point (x, y) is in Quadrant II.

Is the product xy positive or negative?

Answer: _____

The point (x, y) is in Quadrant III.

Is the product xy positive or negative?

Answer: _____

For the point (x, y) , the product xy is positive and x is negative.

In which quadrant is the point?

A. Quadrant I

B. Quadrant II

C. Quadrant III

D. Quadrant IV

True or false?

If (x, y) is in Quadrant IV, then the product xy is negative.

A. True

B. False

Find the product of the coordinates of the point $(-3, 5)$.

Product: _____

SET 27 • 6.NS.C.6.b

FORWARD 2

Transformations and the quadrant of the image • Grade 8 • Geometry •
8.G.A.3

The point (2, 5) is rotated 180° about the origin.

What are the coordinates of the image, and in which quadrant is it?

Image: _____

Quadrant: _____

A point in Quadrant II is reflected across the y -axis.
In which quadrant is the image?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

The point $(-3, -4)$ is rotated 90° counterclockwise about the origin.

What are the coordinates of the image?

Image: _____

True or false?

Translating $(1, 2)$ 5 units to the left moves it into Quadrant II.

- A. True
- B. False

The point $(4, -1)$ is reflected across the x-axis.
In which quadrant is the image?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

Reflect points across one or both axes

MAIN	Grade 6 main question	6.NS.C.6.b
BACKWARD 1	Opposite numbers	6.NS.C.6.a
BACKWARD 2	Lines of symmetry	4.G.A.3
BACKWARD 3	Reading coordinates in the first quadrant	5.G.A.2
FORWARD 1	Distance between a point and its reflection	7.NS.A.1.c
FORWARD 2	Reflections and rotations with coordinates	8.G.A.3

SET 28 • 6.NS.C.6.b

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.b

The point $(3, -5)$ is reflected across the x -axis.
What are the coordinates of its image?

Image: _____

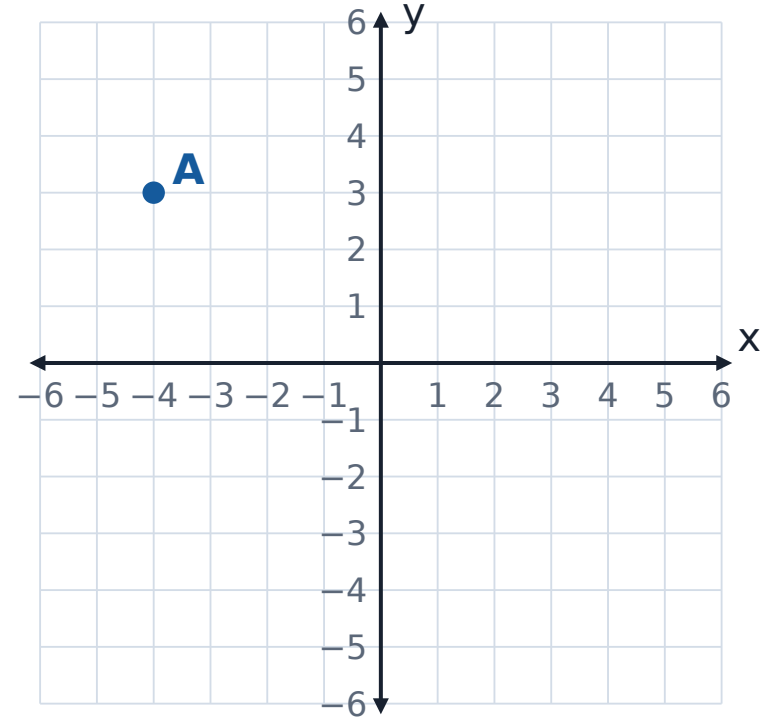
The point $(-2, 6)$ is reflected across the y -axis.
What are the coordinates of its image?

Image: _____

Point A is reflected across the y-axis.

What are the coordinates of its image?

Image: _____



Which point is the reflection of $(-7, -2)$ across the x-axis?

- A.** $(7, 2)$
- B.** $(-2, -7)$
- C.** $(-7, 2)$
- D.** $(7, -2)$

Point B(5, 1) is reflected across the y-axis and then across the x-axis.

What are the coordinates of the final image?

Image: _____

SET 28 • 6.NS.C.6.b

BACKWARD 1

Opposite numbers • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 6?

Opposite: _____

True or false?

The opposite of -9 is 9 .

A. True

B. False

Which pair of numbers are opposites?

- A.** 0 and 4
- B.** -4 and -4
- C.** 4 and $\frac{1}{4}$
- D.** -4 and 4

What is the opposite of -2.5 ?

Opposite: _____

True or false?

$$-(-3) = -3$$

A. True

B. False

SET 28 • 6.NS.C.6.b

BACKWARD 2

Lines of symmetry • Grade 4 • Geometry • 4.G.A.3

True or false?

A line of symmetry divides a figure into two matching halves.

A. True

B. False

How many lines of symmetry does a square have?

A. 0

B. 1

C. 2

D. 4

True or false?

The dashed line is a line of symmetry of the rectangle.

A. True

B. False



Which letter has a vertical line of symmetry?

A. F

B. A

C. J

D. R

How many lines of symmetry does a rectangle that is not a square have?

Lines: _____

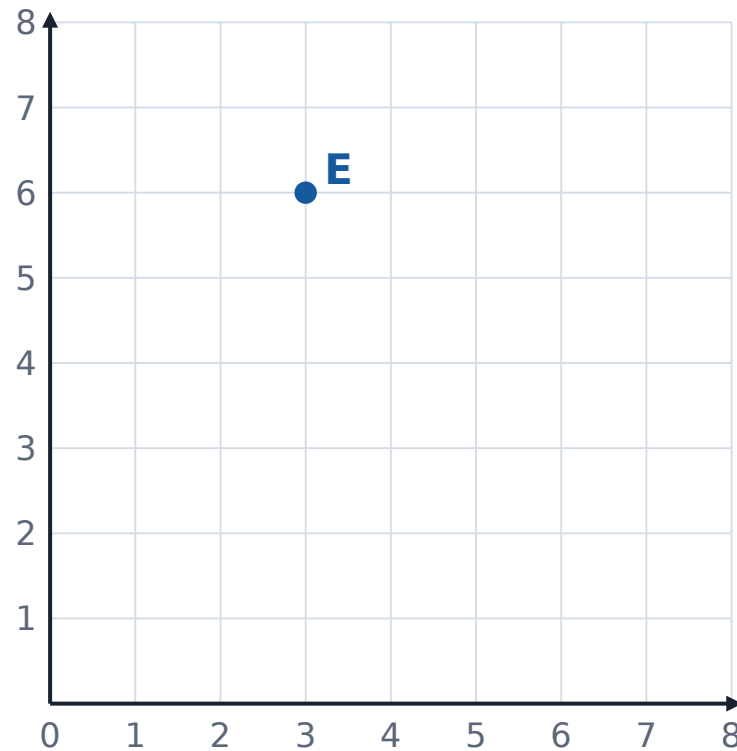
SET 28 • 6.NS.C.6.b

BACKWARD 3

Reading coordinates in the first quadrant • Grade 5 • Geometry • 5.G.A.2

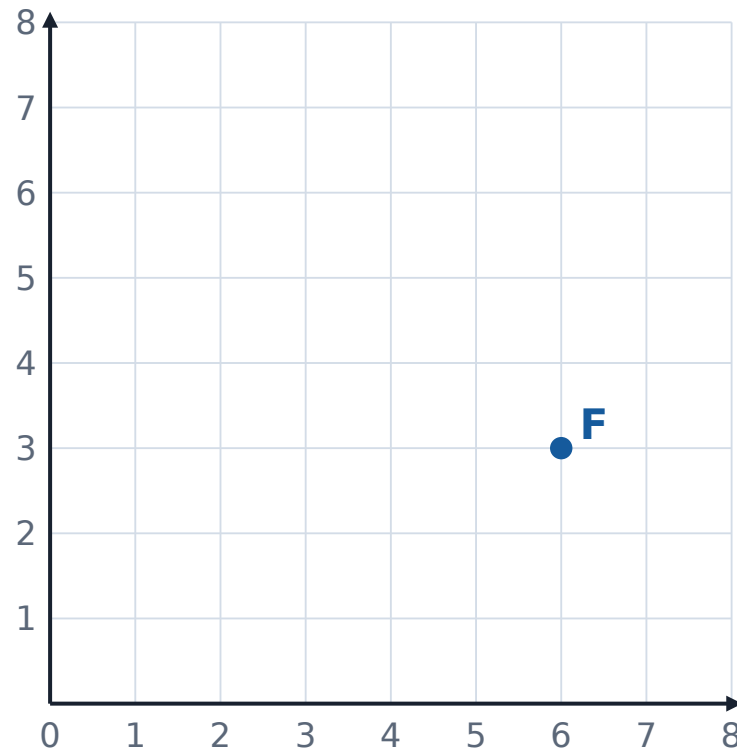
What are the coordinates of point E?

E = _____



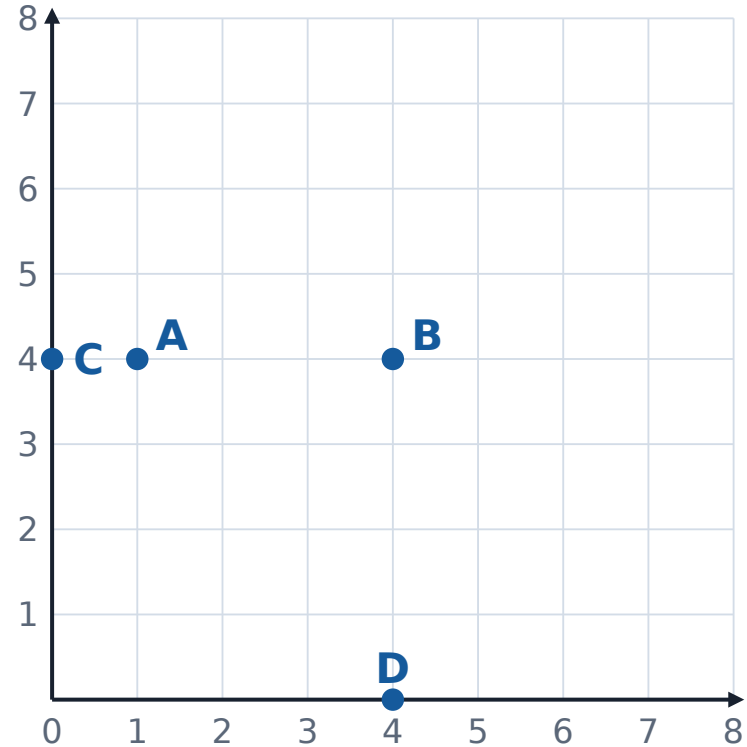
What are the coordinates of point F?

F = _____



Which point is located at (4, 0)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D

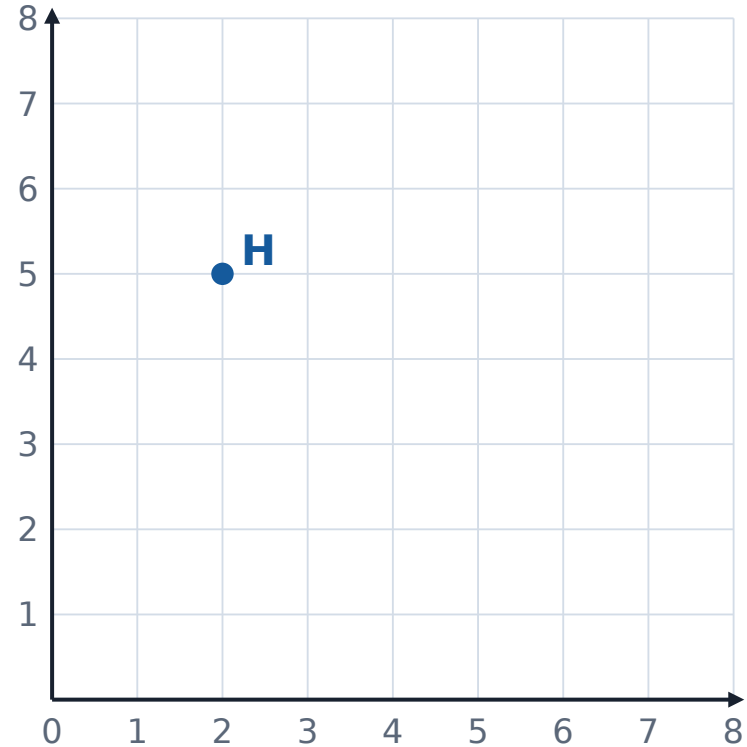


True or false?

Point H is located at (5, 2).

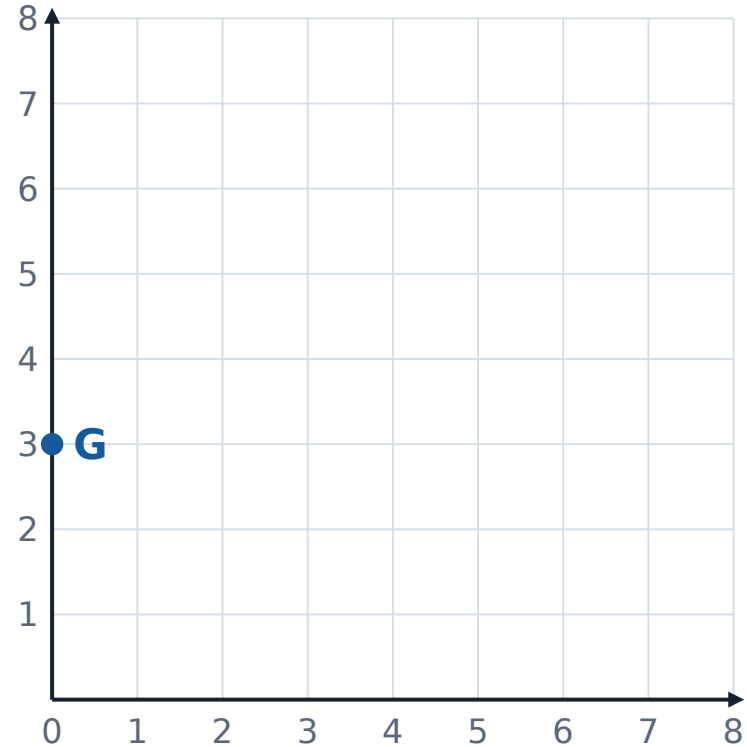
A. True

B. False



What are the coordinates of point G?

G = _____



SET 28 • 6.NS.C.6.b

FORWARD 1

Distance between a point and its reflection • Grade 7 • The Number System •
7.NS.A.1.c

The point $(3, -4.5)$ is reflected across the x-axis.
How far apart are the point and its image?

Distance: _____

The point $(-6.5, 2)$ is reflected across the y -axis.
How far apart are the point and its image?

Distance: _____

The point $(1, -2\frac{1}{2})$ is reflected across the x-axis.

How far apart are the point and its image?

A. -5

B. 0

C. $2\frac{1}{2}$

D. 5

A point in Quadrant I and its reflection across the y-axis are 11 units apart. The point's y-coordinate is 7.

What is the point's x-coordinate?

X = _____

True or false?

The point $(-4, 3)$ and its reflection across the x-axis are 6 units apart.

A. True

B. False

SET 28 • 6.NS.C.6.b

FORWARD 2

Reflections and rotations with coordinates • Grade 8 • Geometry • 8.G.A.3

A triangle has a vertex at $(2, -3)$. The triangle is reflected across the y -axis.

What are the coordinates of the image of that vertex?

Image: _____

Which transformation is described by $(x, y) \rightarrow (x, -y)$?

- A.** A rotation of 180°
- B.** A reflection across the x-axis
- C.** A translation
- D.** A reflection across the y-axis

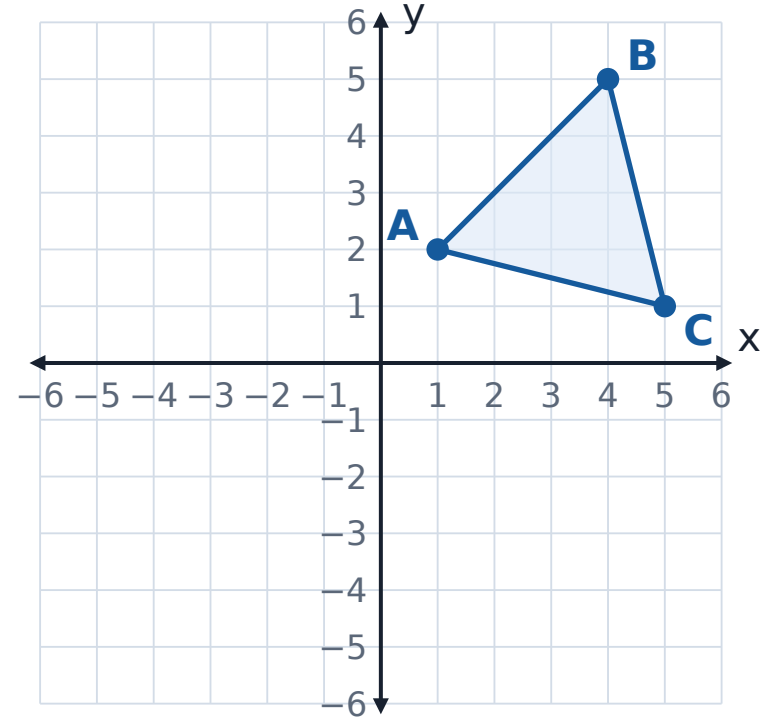
The point $(4, 1)$ is rotated 180° about the origin.
What are the coordinates of its image?

Image: _____

Triangle ABC is reflected
across the x-axis.

What are the coordinates of
 B' ?

$B' =$ _____



True or false?

Reflecting $(-3, 5)$ across the y -axis gives $(3, 5)$.

A. True

B. False

Name the rational number at a point on a number line

MAIN	Grade 6 main question	6.NS.C.6.c
BACKWARD 1	Fractions on a number line	3.NF.A.2
BACKWARD 2	Decimals on a number line	4.NF.C.6
BACKWARD 3	Opposites on a number line	6.NS.C.6.a
FORWARD 1	Addition shown on a number line	7.NS.A.1.b
FORWARD 2	Approximate irrational numbers on a number line	8.NS.A.2

SET 29 • 6.NS.C.6.c

MAIN

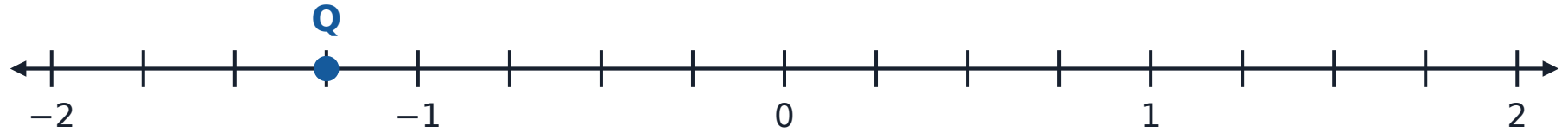
Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

What number does point P represent?



P = _____

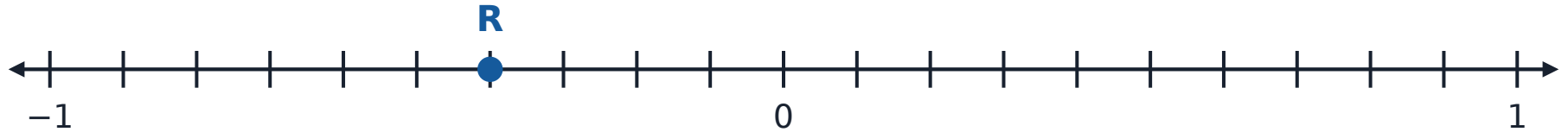
Write the number at point Q as a mixed number.



Q = _____

True or false?

Point R is located at -0.4 .



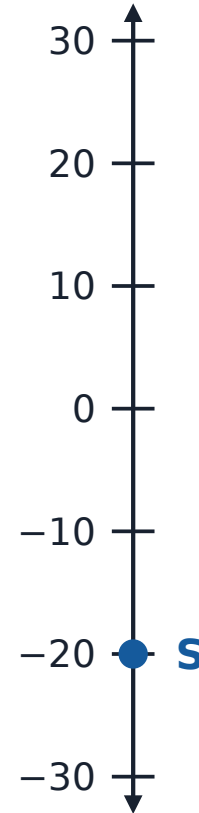
A. True

B. False

The number line shows
elevations in meters.

What elevation does point S
show?

S = _____



Write the number at point T as a fraction.



T = _____

SET 29 • 6.NS.C.6.c

BACKWARD 1

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction does point A represent?



A = _____

What fraction does point B represent?



B = _____

True or false?

Point C is located at $\frac{5}{6}$.



A. True

B. False

The space from 0 to 1 is split into equal parts.
How many equal parts are there?



A. 7

B. 8

C. 9

D. 10

Write the number at point D as a fraction.



D = _____

SET 29 • 6.NS.C.6.c

BACKWARD 2

Decimals on a number line • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

What decimal does point A represent?



A = _____

What decimal does point B represent?



B = _____

True or false?

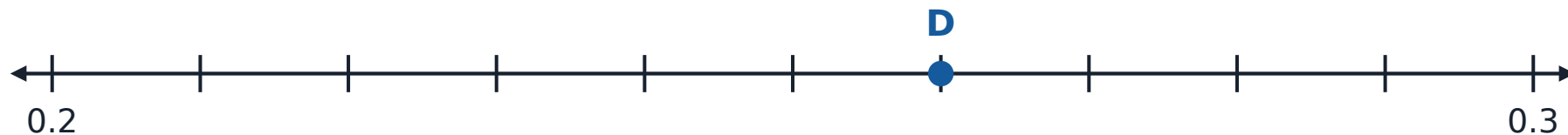
Point C is located at 0.3.



A. True

B. False

What decimal does point D represent?



A. 0.206

B. 0.26

C. 0.62

D. 2.6

What decimal does point E represent?



E = _____

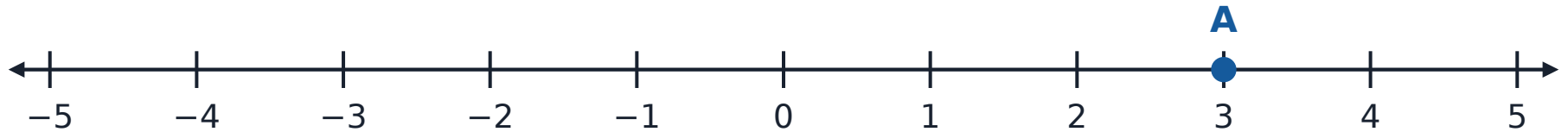
SET 29 • 6.NS.C.6.c

BACKWARD 3

Opposites on a number line • Grade 6 • The Number System • 6.NS.C.6.a

Point B (not shown) is the opposite of point A.

What number is at point B?



B = _____

True or false?

–2.5 and 2.5 are the same distance from 0.

A. True

B. False

What is the opposite of $1\frac{3}{4}$?

A. $-1\frac{3}{4}$

B. $-\frac{4}{7}$

C. $\frac{4}{7}$

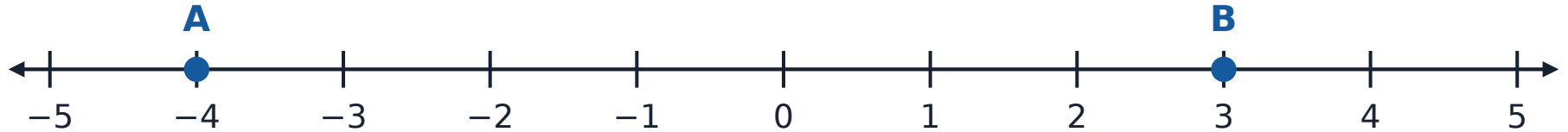
D. $1\frac{3}{4}$

What is the opposite of -0.8 ?

Opposite: _____

True or false?

Points A and B represent opposite numbers.



A. True

B. False

SET 29 • 6.NS.C.6.c

FORWARD 1

Addition shown on a number line • Grade 7 • The Number System • 7.NS.A.1.b

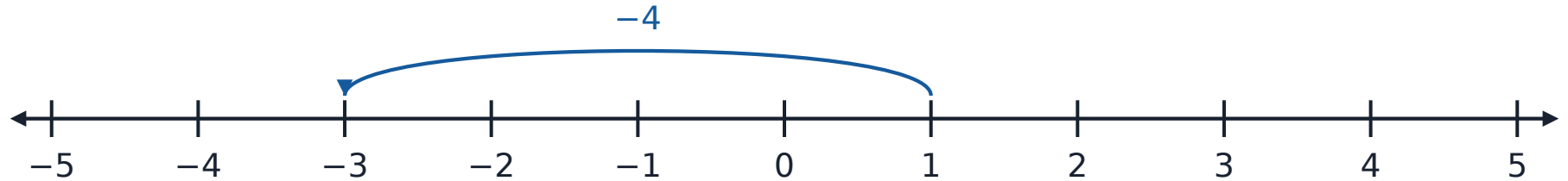
Start at -2.5 on a number line and move 4 units to the right.

Where do you end?

Number: _____

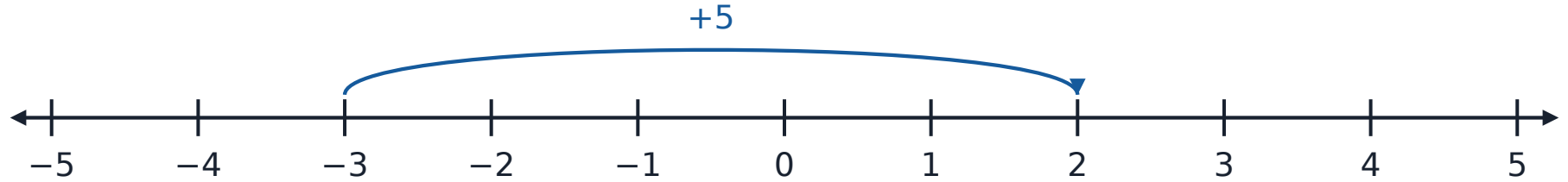
The number line shows $1 + (-4)$.

What is the sum?



Sum: _____

Which expression does the number line show?



A. $5 + 3$

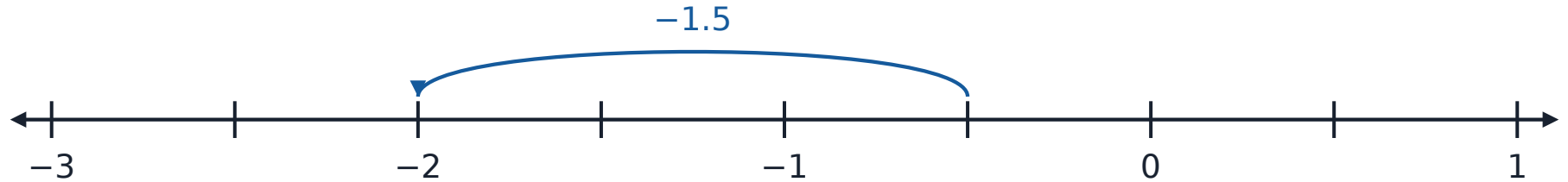
B. $-3 - 5$

C. $-3 + 5$

D. $2 + 5$

The number line shows $-0.5 + (-1.5)$.

What is the sum?



Sum: _____

True or false?

On a number line, $-3 + 7$ is 7 units to the right of -3 .

A. True

B. False

SET 29 • 6.NS.C.6.c

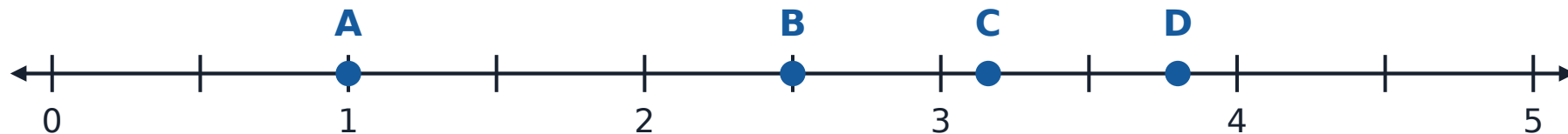
FORWARD 2

Approximate irrational numbers on a number line • Grade 8 • The Number System • 8.NS.A.2

Between which two consecutive whole numbers is $\sqrt{20}$?

Between: _____

Which point best shows $\sqrt{10}$?



A. Point A

B. Point B

C. Point C

D. Point D

Estimate $\sqrt{50}$ to the nearest tenth.

Estimate: _____

Which list is ordered from least to greatest?

A. 2.5, $\sqrt{5}$, $\sqrt{8}$

B. $\sqrt{5}$, 2.5, $\sqrt{8}$

C. $\sqrt{8}$, $\sqrt{5}$, 2.5

D. $\sqrt{5}$, $\sqrt{8}$, 2.5

True or false?

π is between 3.1 and 3.2 on the number line.

A. True

B. False

Plot rational numbers on a horizontal or vertical number line

MAIN	Grade 6 main question	6.NS.C.6.c
BACKWARD 1	Placing fractions on a number line	3.NF.A.2
BACKWARD 2	Placing decimals on a number line	4.NF.C.6
BACKWARD 3	Opposites on a number line	6.NS.C.6.a
FORWARD 1	Represent addition on a number line	7.NS.A.1.b
FORWARD 2	Locate irrational numbers approximately on a number line	8.NS.A.2

SET 30 • 6.NS.C.6.c

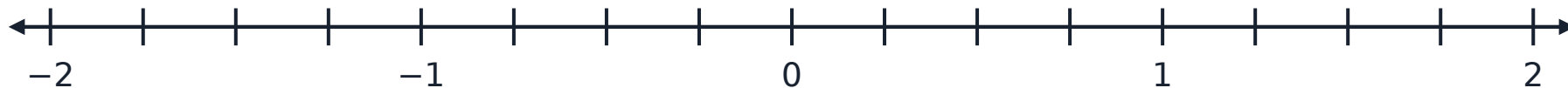
MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

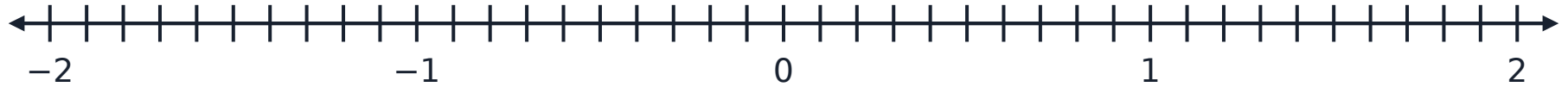
Plot and label point A at -2.5 on the number line.



Plot and label point B at $-\frac{3}{4}$ on the number line.

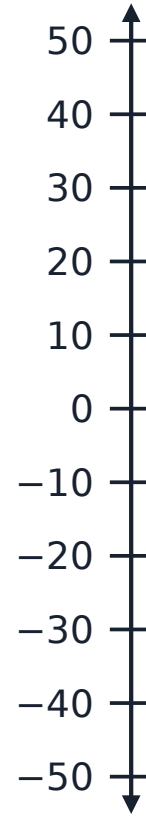


Plot and label -1.2 and its opposite on the number line.

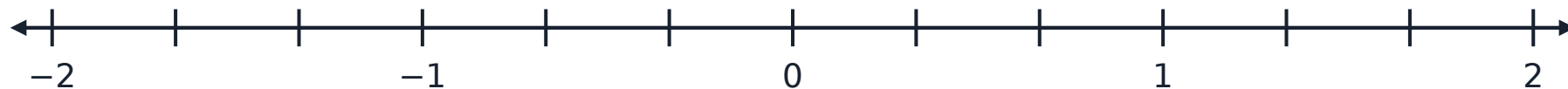


The number line shows
elevations in feet.

Plot and label point C at -35
feet.



Plot and label point D at $1\frac{2}{3}$ on the number line.



SET 30 • 6.NS.C.6.c

BACKWARD 1

Placing fractions on a number line • Grade 3 • Number & Operations–Fractions
• 3.NF.A.2

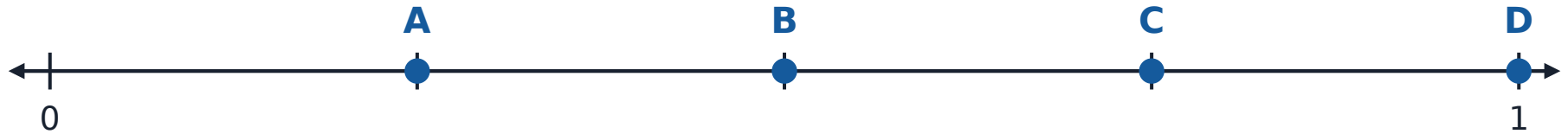
Plot and label $\frac{3}{4}$ on the number line.



Plot and label $\frac{2}{3}$ on the number line.



Which point is at $\frac{1}{2}$?



A. Point A

B. Point B

C. Point C

D. Point D

Plot and label $\frac{5}{6}$ on the number line.



True or false?

To plot $\frac{3}{8}$, count 3 of the 8 equal parts from 0.



A. True

B. False

SET 30 • 6.NS.C.6.c

BACKWARD 2

Placing decimals on a number line • Grade 4 • Number & Operations–Fractions
• 4.NF.C.6

Plot and label 0.7 on the number line.



Plot and label 2.4 on the number line.



Which point is at 0.6?



A. Point A

B. Point B

C. Point C

D. Point D

Plot and label 0.35 on the number line.



True or false?

On a number line marked in tenths, 0.9 is one tick mark to the left of 1.

A. True

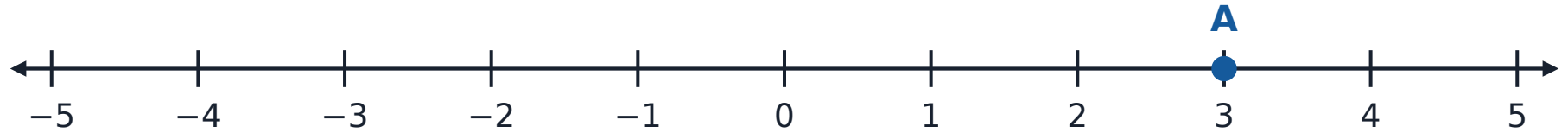
B. False

SET 30 • 6.NS.C.6.c

BACKWARD 3

Opposites on a number line • Grade 6 • The Number System • 6.NS.C.6.a

Point A is at 3. Plot and label its opposite.



Plot and label 2 and -2 on the number line.



True or false?

A number and its opposite are the same distance from 0.

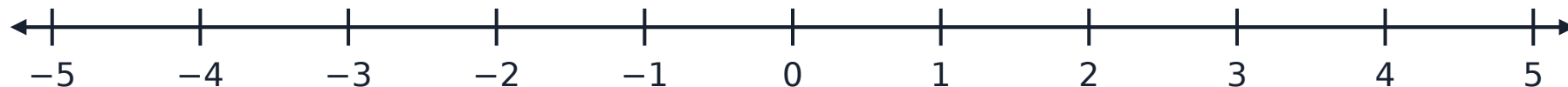
A. True

B. False

Point B is at -1.5 . Where is its opposite?

- A.** At -1.5
- B.** At 1.5
- C.** At 0
- D.** At 0.15

Plot and label the opposite of -4 .



SET 30 • 6.NS.C.6.c

FORWARD 1

Represent addition on a number line • Grade 7 • The Number System •
7.NS.A.1.b

Draw an arrow on the number line to show $-2 + 3.5$.

What is the sum?



Sum: _____

Draw arrows on the number line to show $1.5 + (-3)$.

What is the sum?

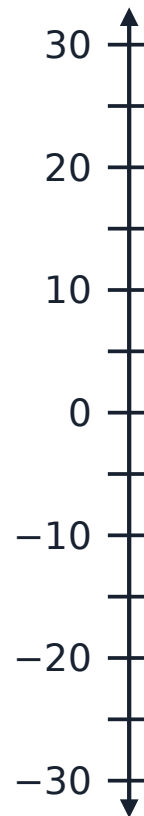


Sum: _____

Draw arrows on the vertical
number line to show
 $-10 + 25$.

What is the sum?

Sum: _____



An arrow on a number line starts at $-\frac{1}{2}$ and moves $\frac{3}{4}$ unit to the right.

Where does it end?

A. $-1\frac{1}{4}$

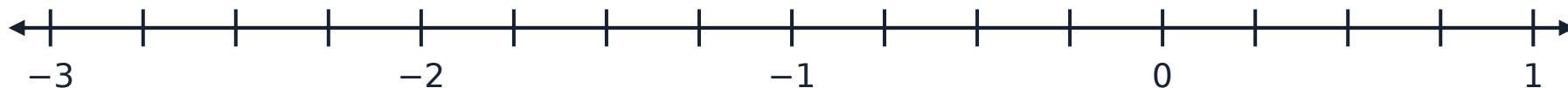
B. $-\frac{1}{4}$

C. $\frac{1}{4}$

D. $\frac{3}{4}$

Draw arrows on the number line to show $-1\frac{1}{4} + (-\frac{3}{4})$.

What is the sum?



Sum: _____

SET 30 • 6.NS.C.6.c

FORWARD 2

Locate irrational numbers approximately on a number line • Grade 8 • The Number System • 8.NS.A.2

Plot $\sqrt{10}$ at its approximate location on the number line.



Plot $\sqrt{2}$ at its approximate location on the number line.



Plot π at its approximate location on the number line.



Plot $-\sqrt{5}$ at its approximate location on the number line.



Plot $\sqrt{30}$ at its approximate location on the number line.



Name the coordinates of points in all four quadrants

MAIN	Grade 6 main question	6.NS.C.6.c
BACKWARD 1	Ordered pairs and the axes	5.G.A.1
BACKWARD 2	Reading points in the first quadrant	5.G.A.2
BACKWARD 3	Quadrants and signs of coordinates	6.NS.C.6.b
FORWARD 1	Interpret a point on the graph of a proportional relationship	7.RP.A.2.d
FORWARD 2	A function as a set of ordered pairs	8.F.A.1

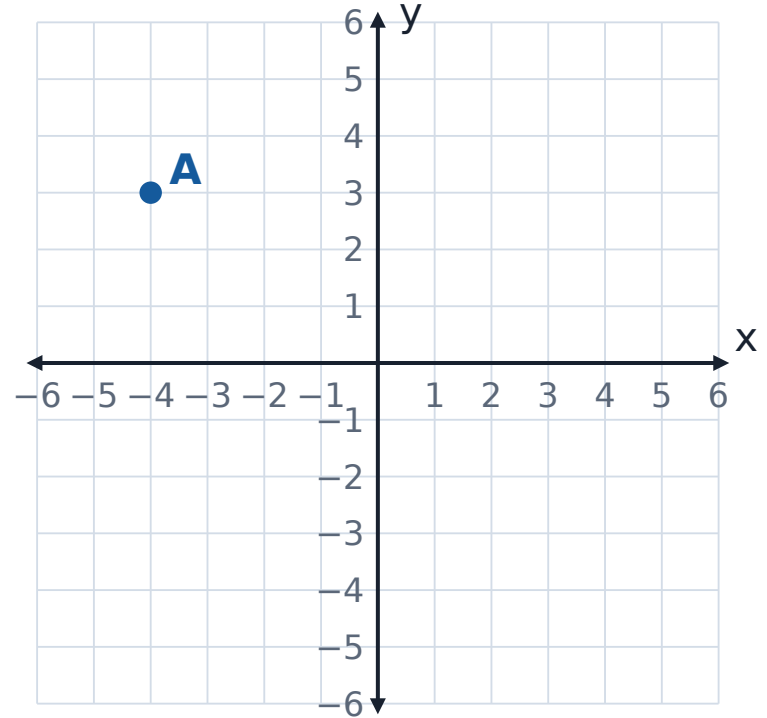
SET 31 • 6.NS.C.6.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

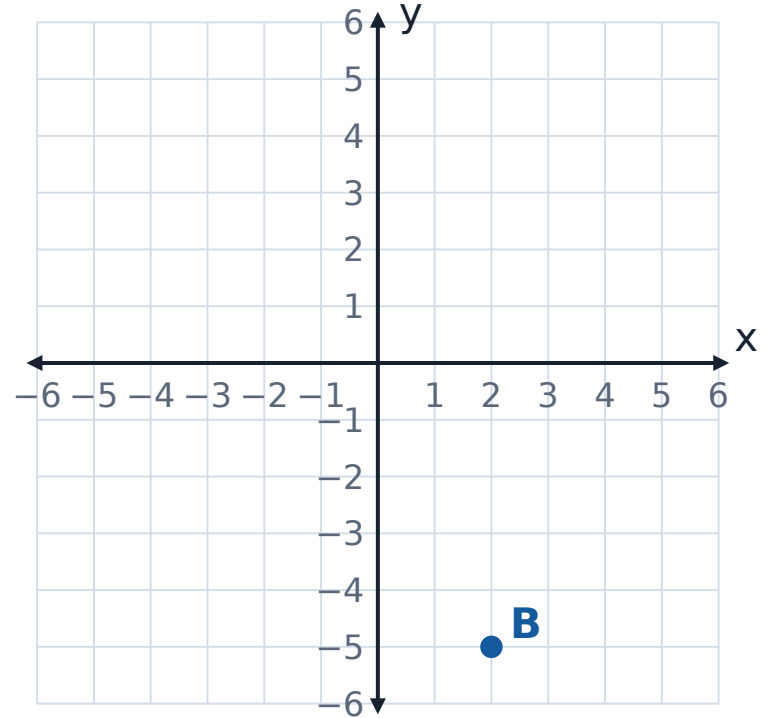
What are the coordinates of point A?

A = _____



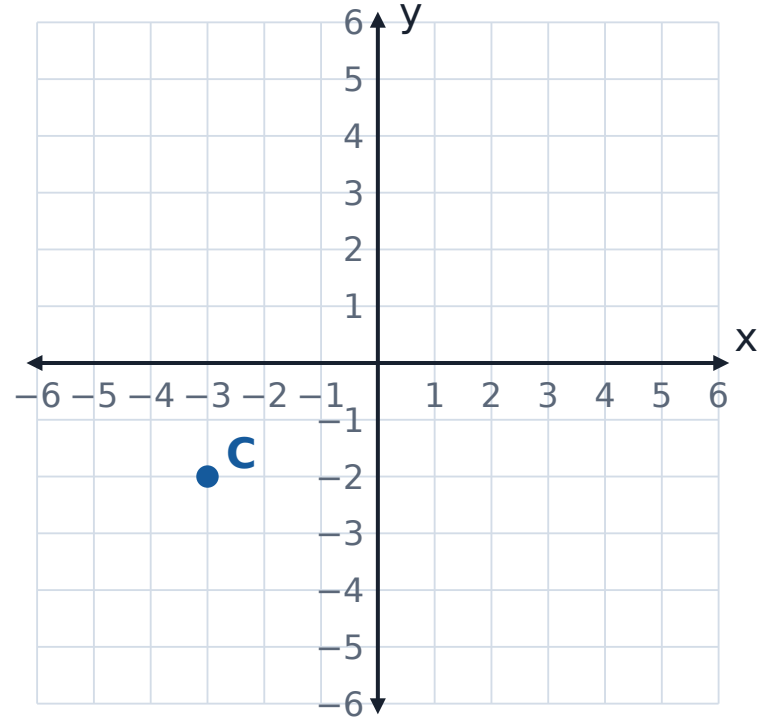
What are the coordinates of point B?

B = _____



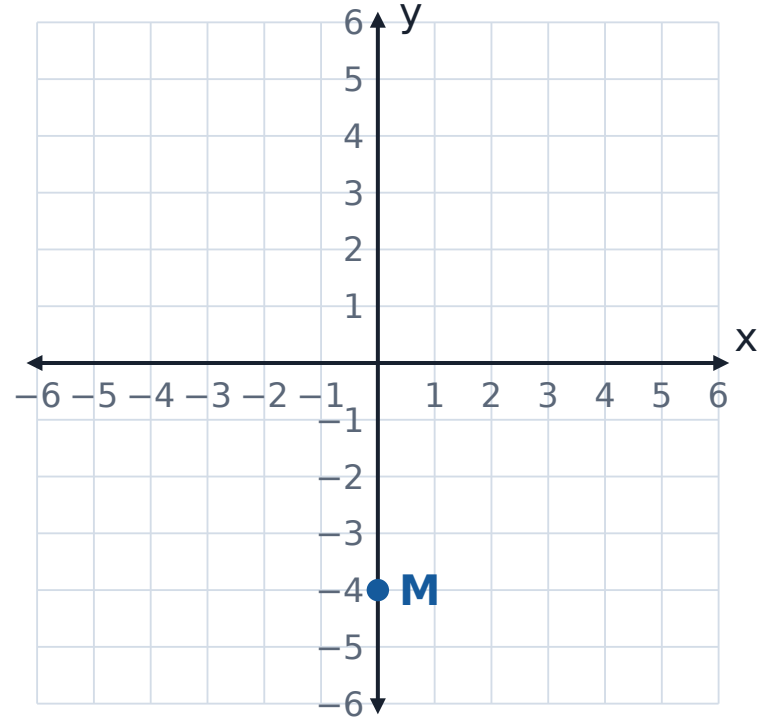
What are the coordinates of point C?

C = _____



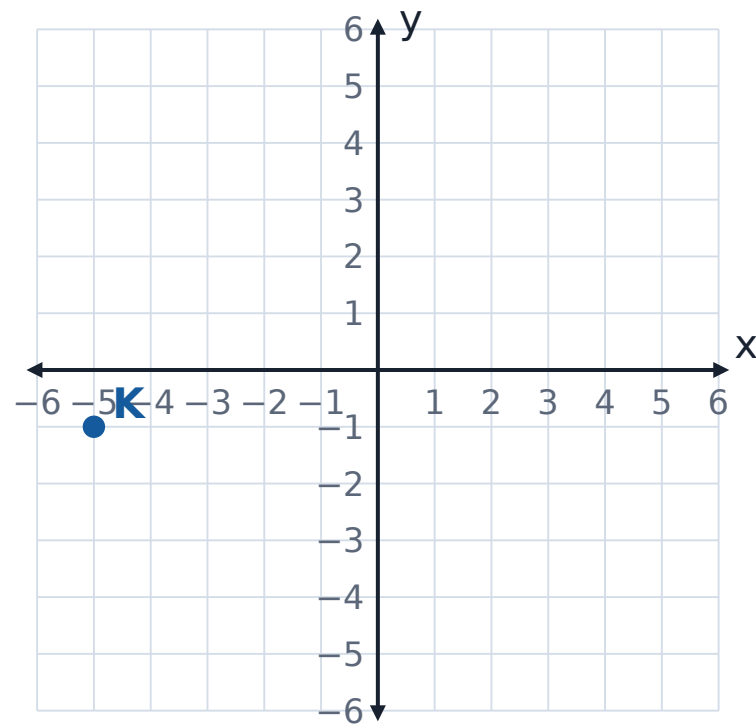
Which ordered pair names point M?

- A.** $(0, 4)$
- B.** $(4, 0)$
- C.** $(0, -4)$
- D.** $(-4, 0)$



What are the coordinates of point K?

K = _____



SET 31 • 6.NS.C.6.c

BACKWARD 1

Ordered pairs and the axes • Grade 5 • Geometry • 5.G.A.1

True or false?

The x-coordinate tells how far to move from the origin along the x-axis.

A. True

B. False

In the ordered pair $(7, 3)$, what is the y-coordinate?

- A.** 3
- B.** 4
- C.** 7
- D.** 10

The x-axis and the y-axis cross at a point.
What are its name and coordinates?

Name: _____

Coordinates: _____

True or false?

$(2, 5)$ and $(5, 2)$ name the same point.

A. True

B. False

In an ordered pair, which number is written first?

- A.** The smaller number
- B.** The x-coordinate
- C.** The y-coordinate
- D.** The larger number

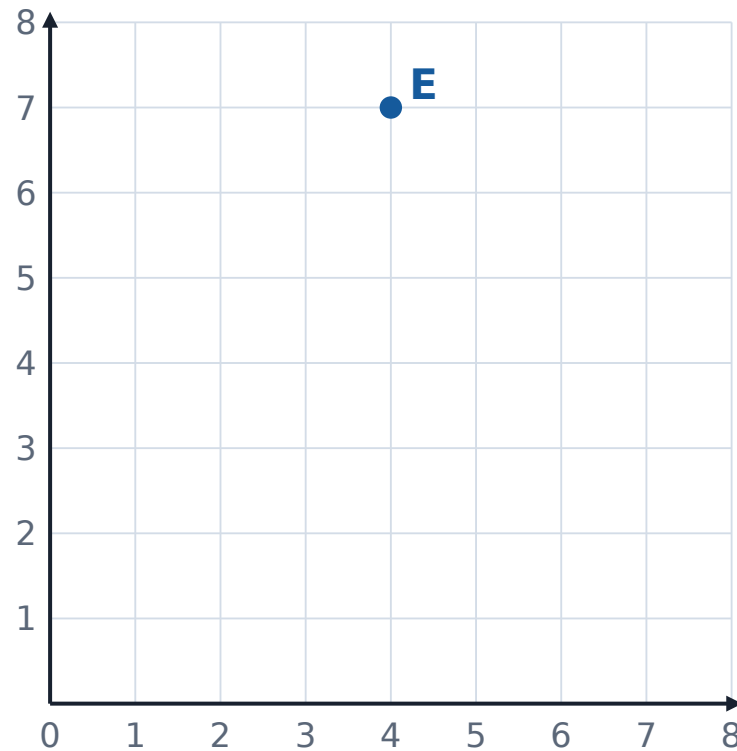
SET 31 • 6.NS.C.6.c

BACKWARD 2

Reading points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

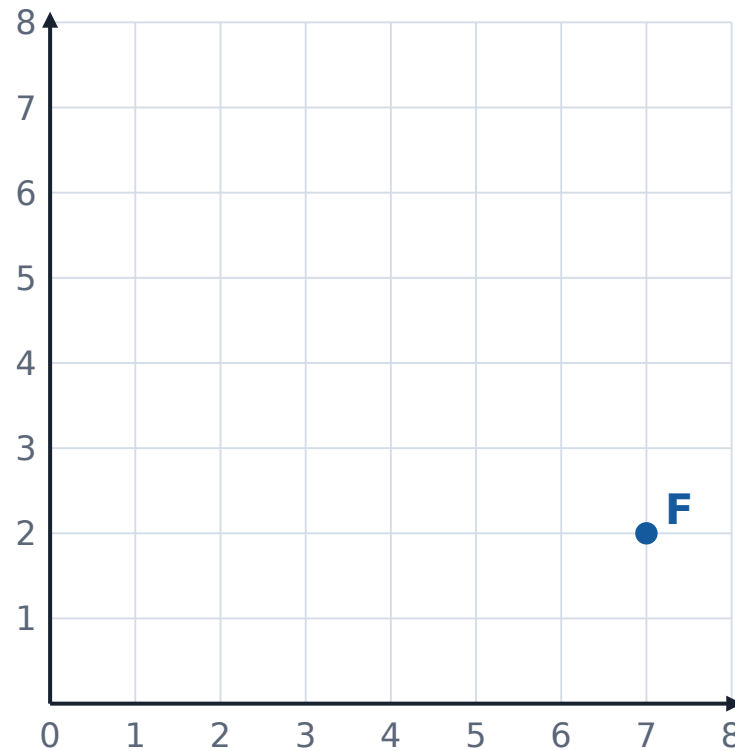
What are the coordinates of point E?

E = _____



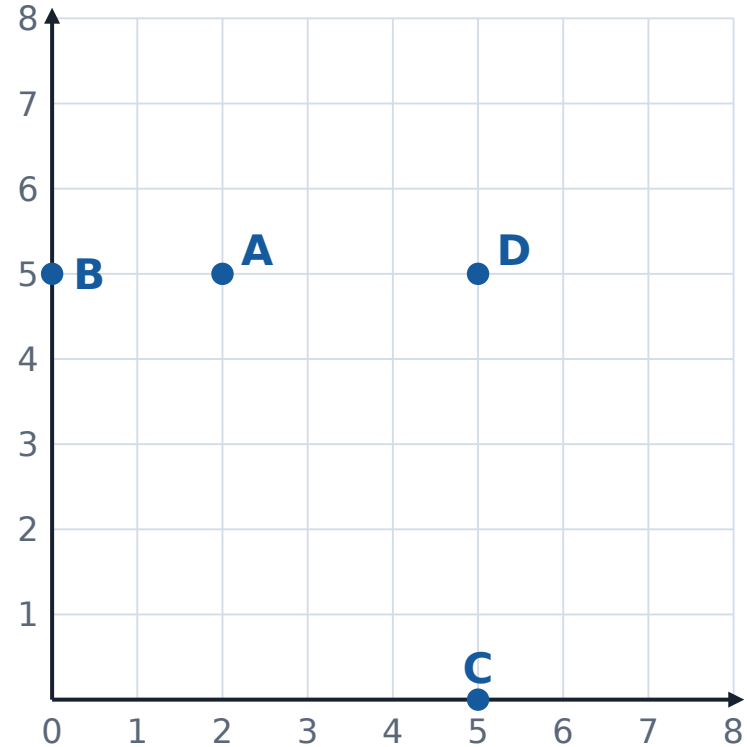
What are the coordinates of point F?

F = _____



Which point is located at (0, 5)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D

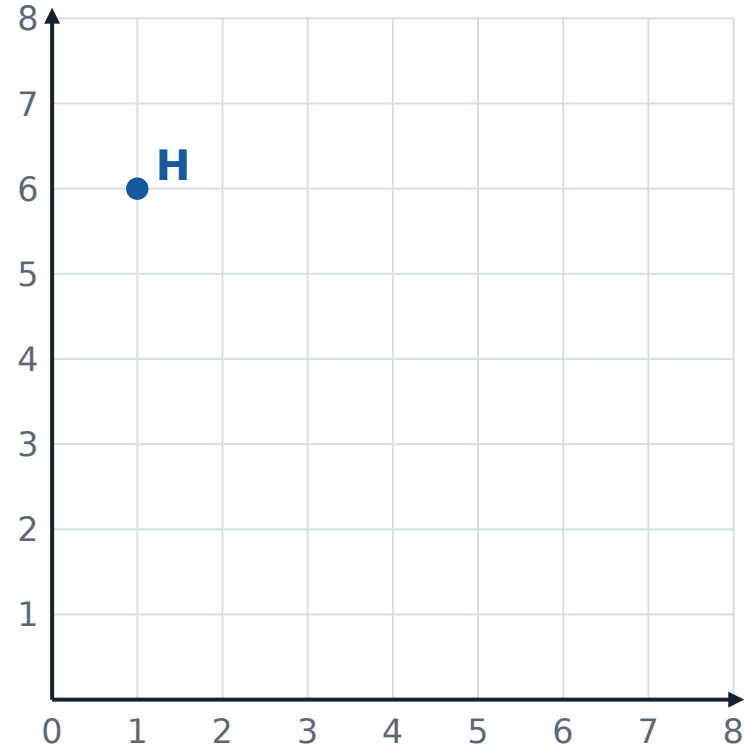


True or false?

Point H is located at (6, 1).

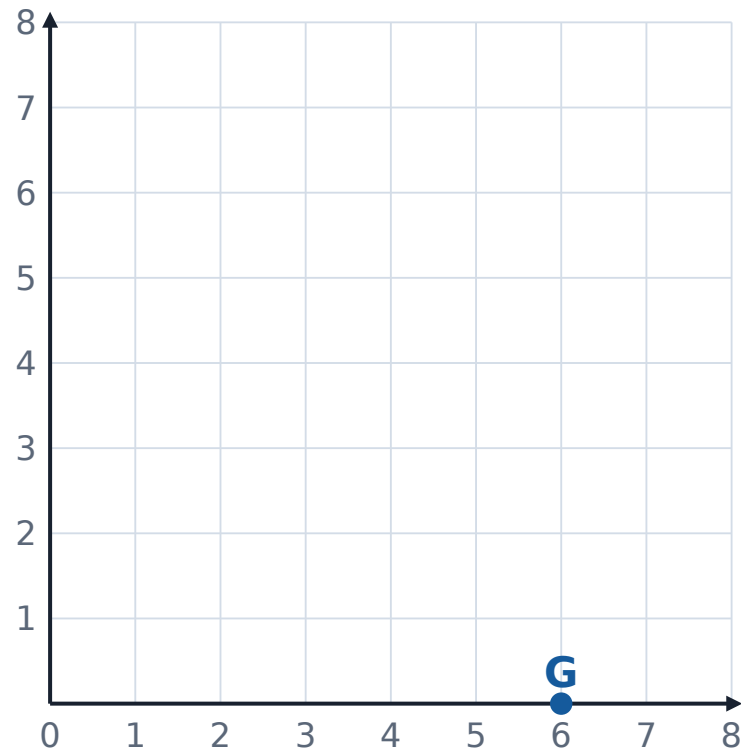
A. True

B. False



What are the coordinates of point G?

G = _____



SET 31 • 6.NS.C.6.c

BACKWARD 3

Quadrants and signs of coordinates • Grade 6 • The Number System •
6.NS.C.6.b

In which quadrant are both coordinates negative?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

In which quadrant is $(5, -2)$?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

True or false?

$(-3, 4)$ is in Quadrant II.

A. True

B. False

A point is in Quadrant IV.

Write "positive" or "negative" for each coordinate.

x-coordinate: _____

y-coordinate: _____

True or false?

The point $(0, -6)$ is in Quadrant III.

A. True

B. False

SET 31 • 6.NS.C.6.c

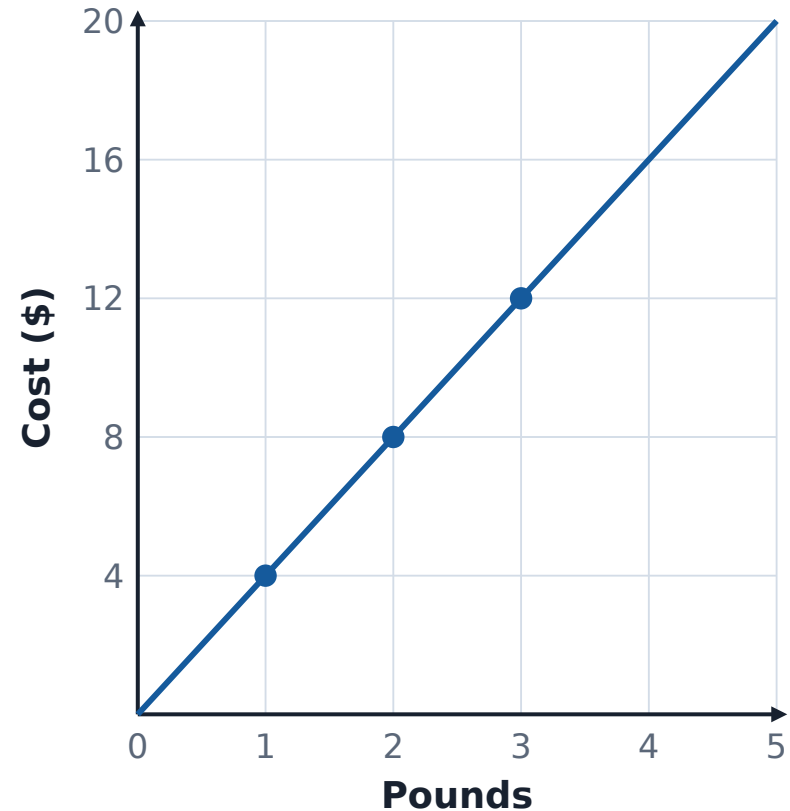
FORWARD 1

Interpret a point on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph shows the cost of apples.

What does the point $(2, 8)$ represent?

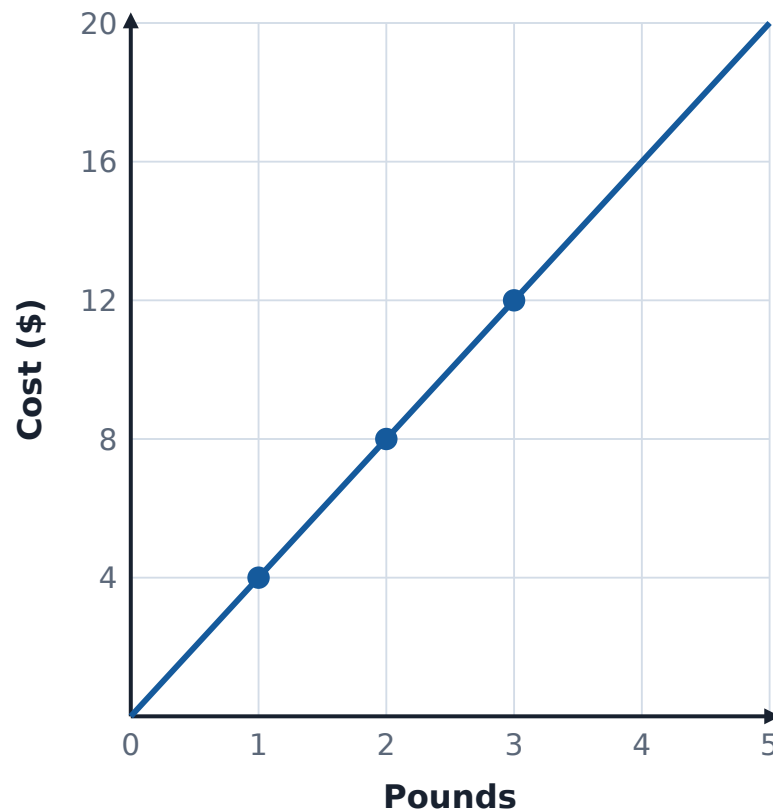
Meaning: _____



The graph shows the cost of apples.

What does the point (1, 4) represent?

- A.** 4 pounds cost \$1.
- B.** 1 pound costs \$4.
- C.** \$1 buys 4 pounds.
- D.** The cost starts at \$4.



True or false?

On the graph of a proportional relationship between pounds and cost, the point $(0, 0)$ means 0 pounds cost \$0.

- A.** True
- B.** False

The graph of a proportional relationship passes through $(1, r)$ and $(5, 30)$.

What is r ?

$r =$ _____

A graph shows the distance y in miles after x hours. It passes through $(3, 150)$.

What does this point mean?

- A.** In 3 hours, the distance is 150 miles.
- B.** The speed is 150 miles per hour.
- C.** In 150 hours, the distance is 3 miles.
- D.** The speed is 3 miles per hour.

SET 31 • 6.NS.C.6.c

FORWARD 2

A function as a set of ordered pairs • Grade 8 • Functions • 8.F.A.1

True or false?

The set of ordered pairs $(1, 2)$, $(2, 4)$, $(3, 6)$, $(4, 8)$ represents a function.

- A. True
- B. False

Which set of ordered pairs does NOT represent a function?

A. $(-1, 3), (0, 5), (1, 7)$

B. $(1, 4), (2, 4), (3, 4)$

C. $(0, 0), (1, 1), (2, 2)$

D. $(2, 5), (2, 7), (3, 9)$

Does the set $(4, 1), (5, 2), (4, 3)$ represent a function?
Write yes or no.

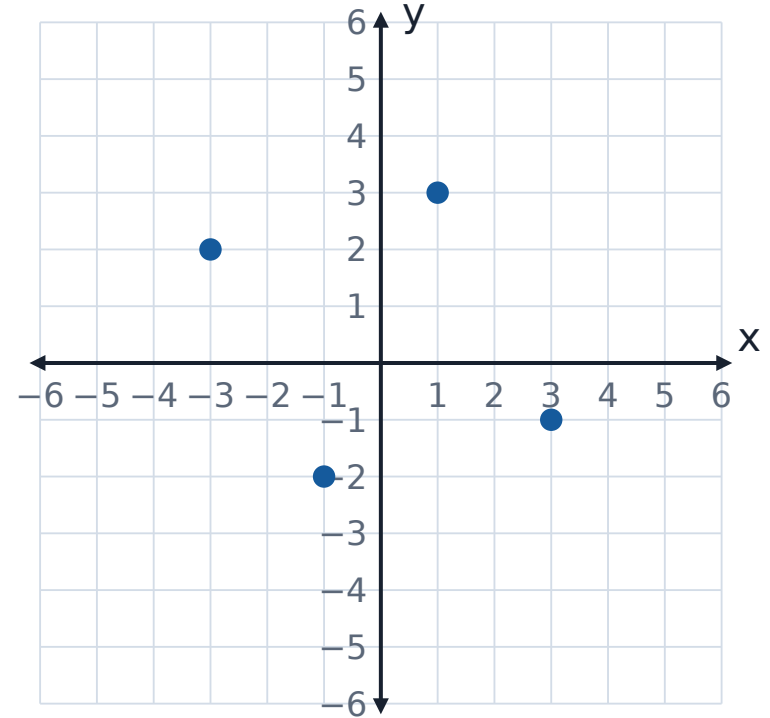
Answer: _____

True or false?

The graph represents a function.

A. True

B. False



A function assigns to each input ____.

- A.** at least two outputs
- B.** no output
- C.** the same output as the input
- D.** exactly one output

Plot points in all four quadrants

MAIN

Grade 6 main question

6.NS.C.6.c

BACKWARD 1

Which direction to move first

5.G.A.1

BACKWARD 2

Plotting points in the first quadrant

5.G.A.2

BACKWARD 3

Direction from the sign of a coordinate

6.NS.C.6.b

FORWARD 1

Graph pairs to decide whether a relationship is proportional

7.RP.A.2.a

FORWARD 2

Graph proportional relationships

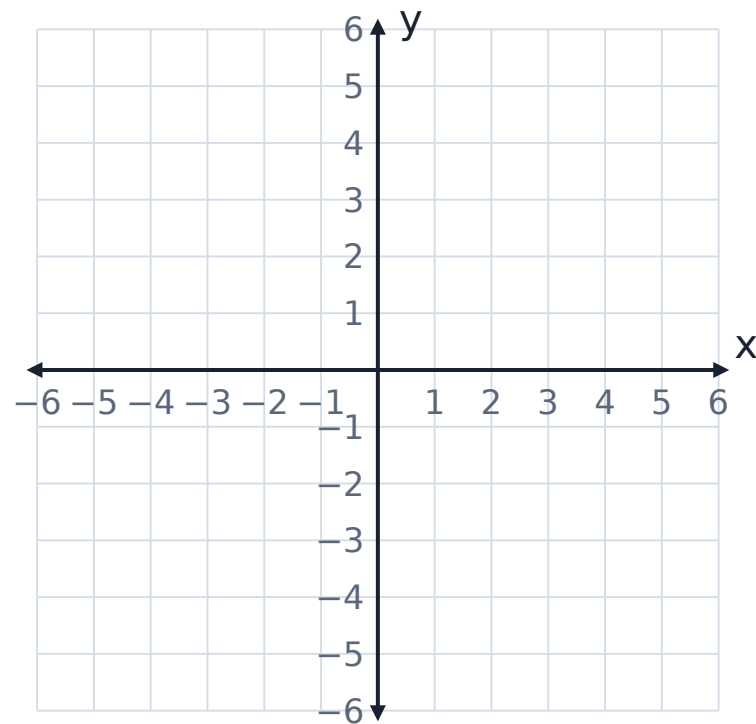
8.EE.B.5

SET 32 • 6.NS.C.6.c

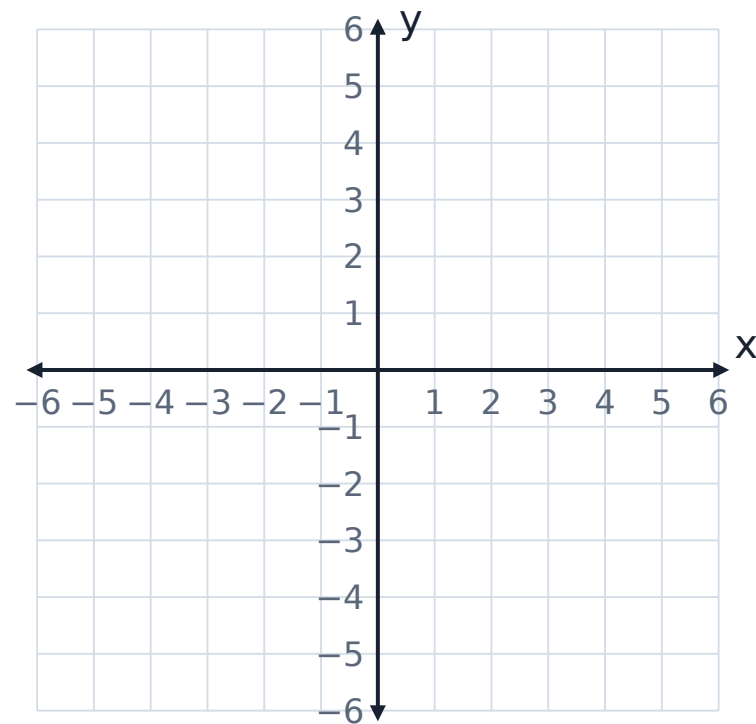
MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

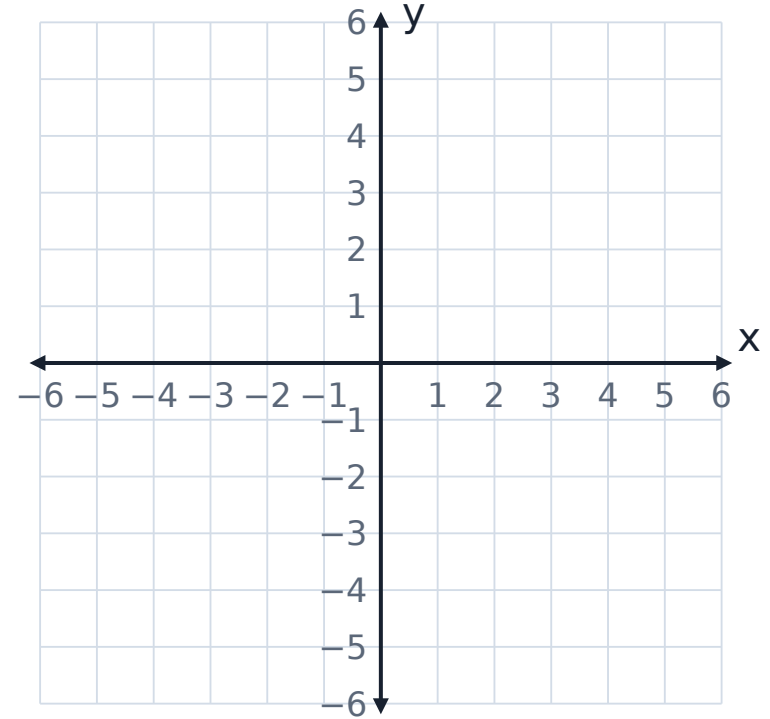
Plot and label point $P(-3, 4)$
and point $Q(2, -5)$.



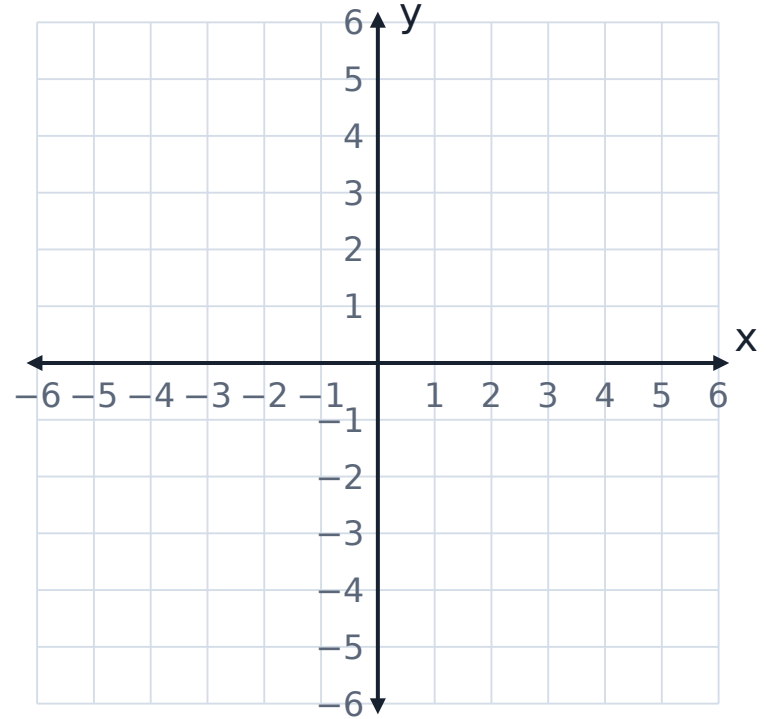
Plot and label point $R(-5, -2)$
and point $S(4, 1)$.



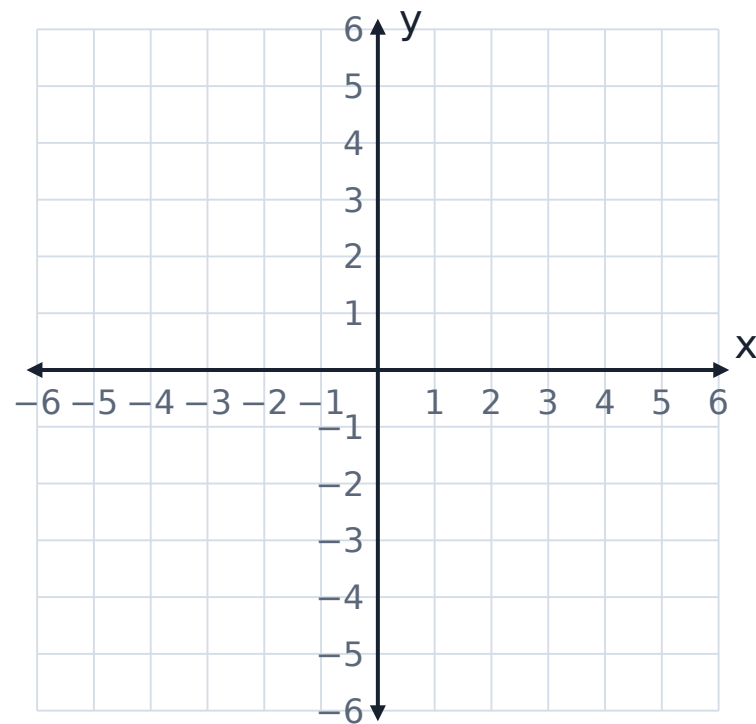
Plot and label point $T(0, -3)$
and point $U(-4, 0)$.



Plot and label point $V(-2.5, 3)$ and point $W(1.5, -2)$.



Plot and label point $X(-6, -6)$
and point $Y(5, -4)$.



SET 32 • 6.NS.C.6.c

BACKWARD 1

Which direction to move first • Grade 5 • Geometry • 5.G.A.1

True or false?

To plot $(6, 1)$, first move 6 units along the x-axis.

A. True

B. False

To plot (2, 7) from the origin, what do you do first?

- A.** Move 7 units up
- B.** Move 2 units right
- C.** Move 2 units up
- D.** Move 7 units right

True or false?

To plot $(0, 5)$, you do not move left or right.

A. True

B. False

To plot (4, 3), how many units up do you move?

Units: _____

Which ordered pair is 3 units right and 6 units up from the origin?

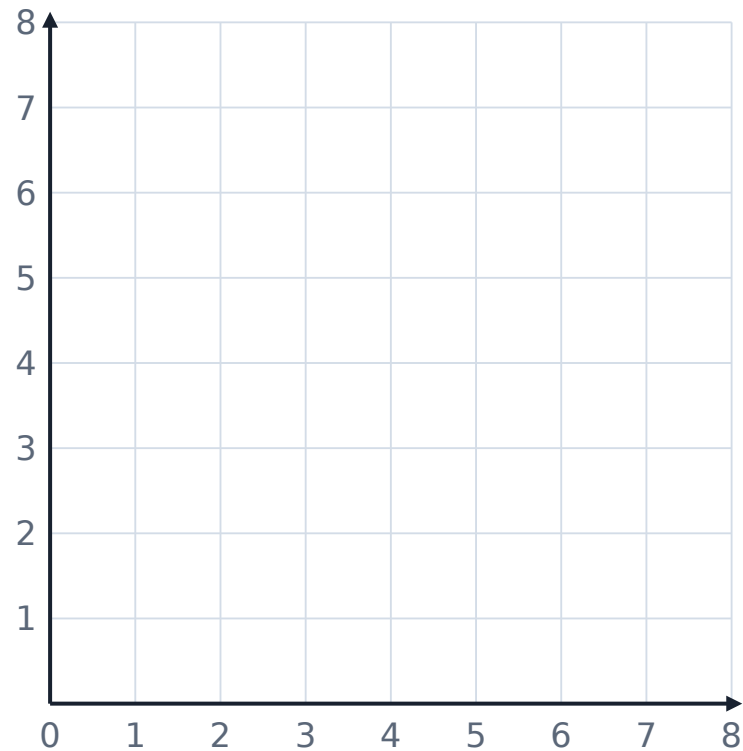
- A.** (3, 3)
- B.** (3, 6)
- C.** (6, 3)
- D.** (6, 6)

SET 32 • 6.NS.C.6.c

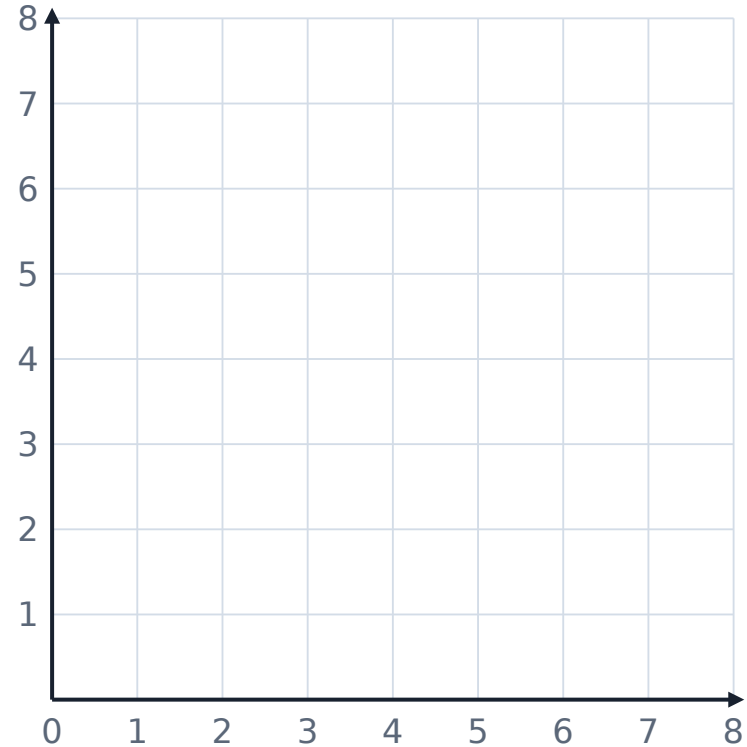
BACKWARD 2

Plotting points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

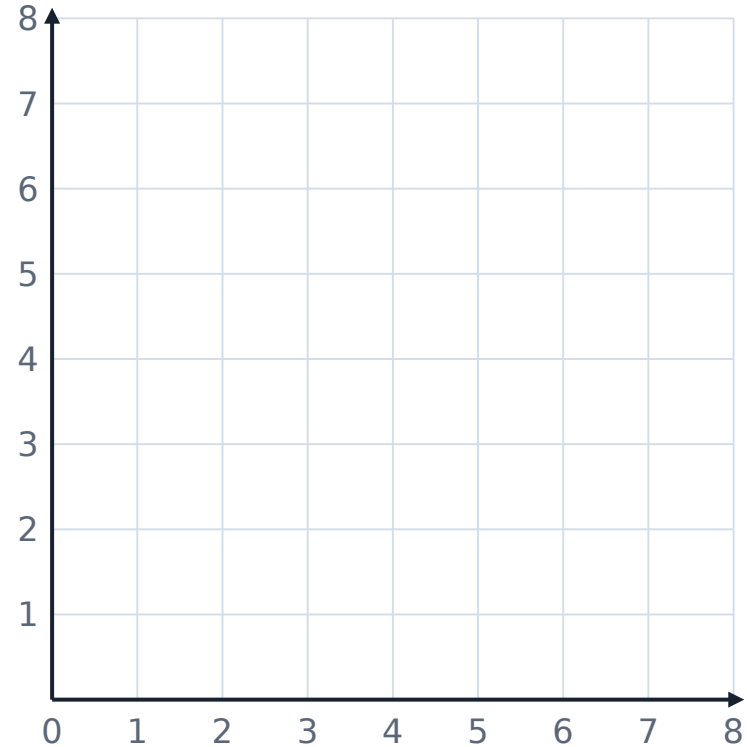
Plot and label point A at (2, 6).



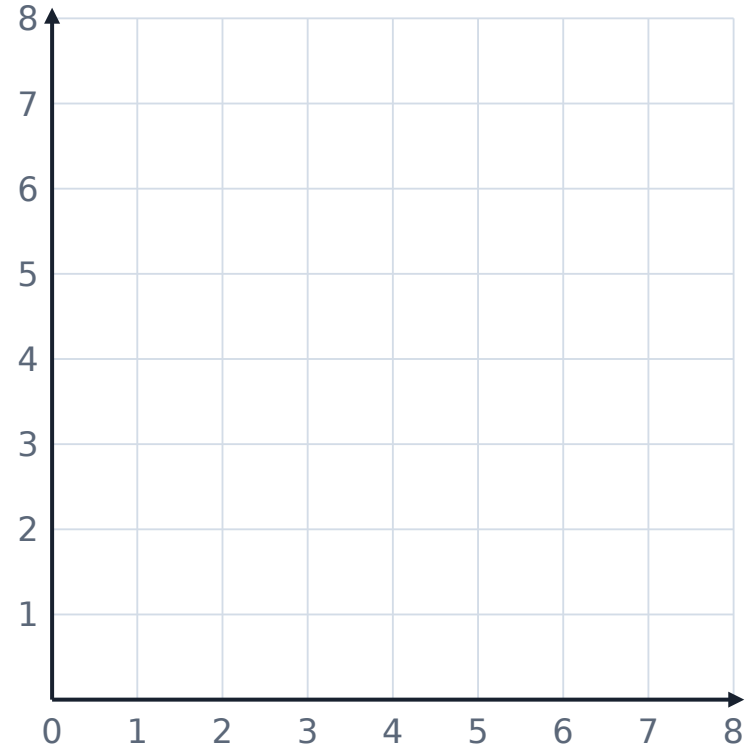
Plot and label point B at (5, 1).



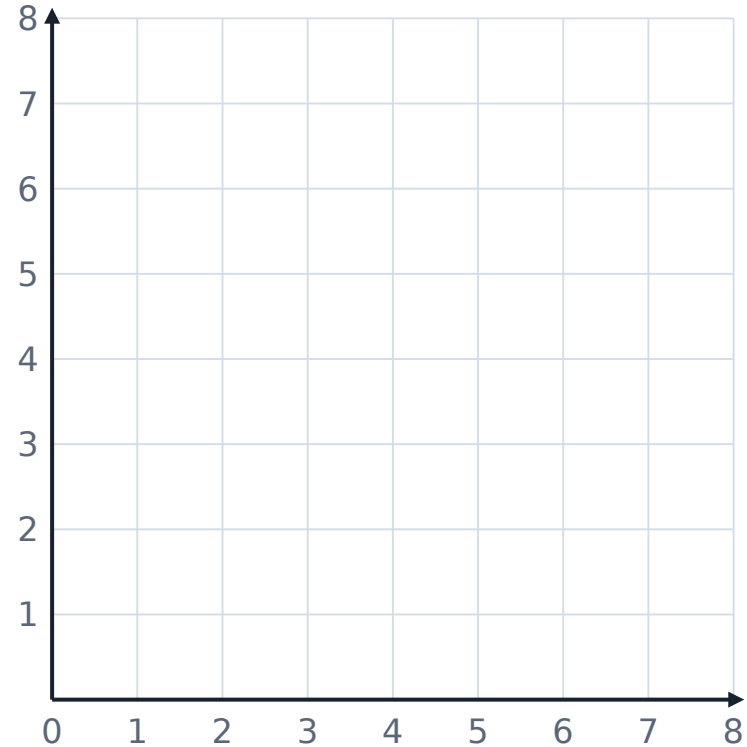
Plot and label point C at (0, 7).



Plot and label point D at (7, 7).



Plot and label point E at (5, 0).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • 6.NS.C.6.c

BACKWARD 3

Direction from the sign of a coordinate • Grade 6 • The Number System •
6.NS.C.6.b

True or false?

To plot a point with a negative x-coordinate, you move left from the origin.

- A.** True
- B.** False

To plot $(-2, -5)$, which directions do you move?

- A.** Right 2, down 5
- B.** Right 2, up 5
- C.** Left 2, down 5
- D.** Left 5, down 2

True or false?

To plot $(3, -4)$, you move up 4 units.

- A. True
- B. False

To plot $(-6, 1)$, do you move left or right?

Answer: _____

Which point is left of the y -axis and above the x -axis?

- A.** $(-3, 2)$
- B.** $(3, -2)$
- C.** $(3, 2)$
- D.** $(-3, -2)$

SET 32 • 6.NS.C.6.c

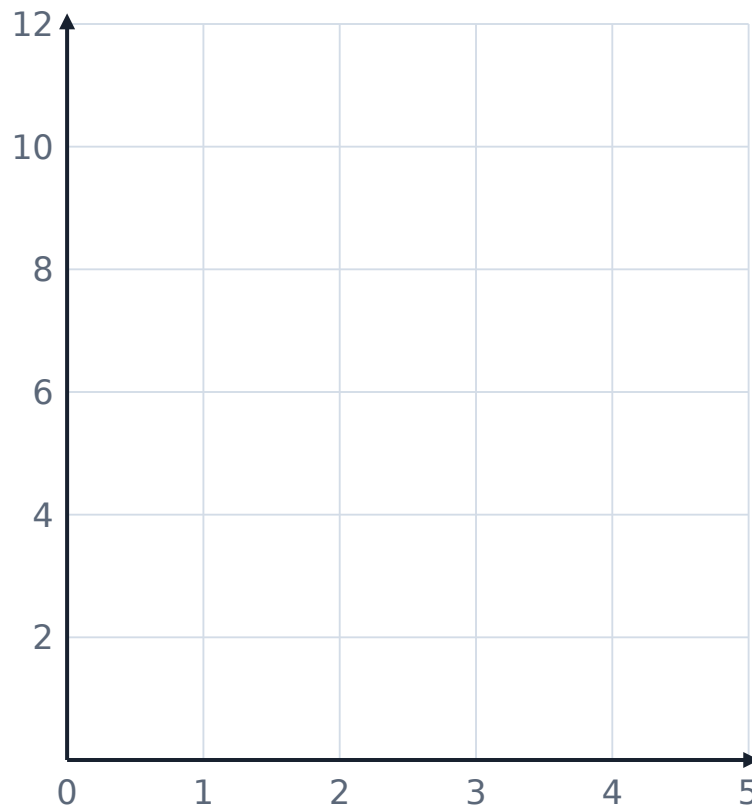
FORWARD 1

Graph pairs to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Graph the pairs (1, 3), (2, 6),
and (3, 9).

Do they lie on a straight line
through the origin?

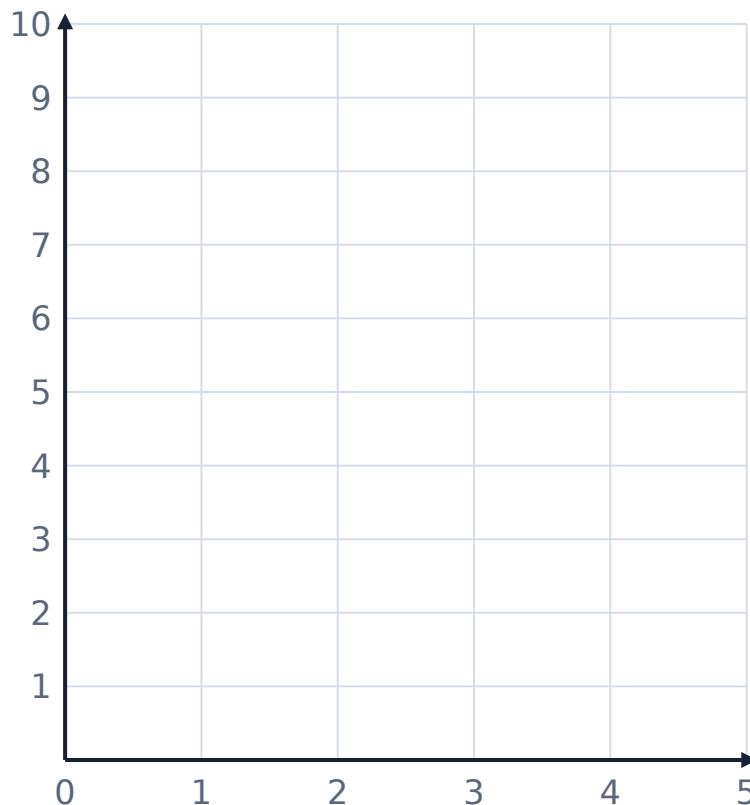
Answer: _____



Graph the pairs $(1, 3)$, $(2, 5)$,
and $(3, 7)$.

Is the relationship
proportional?

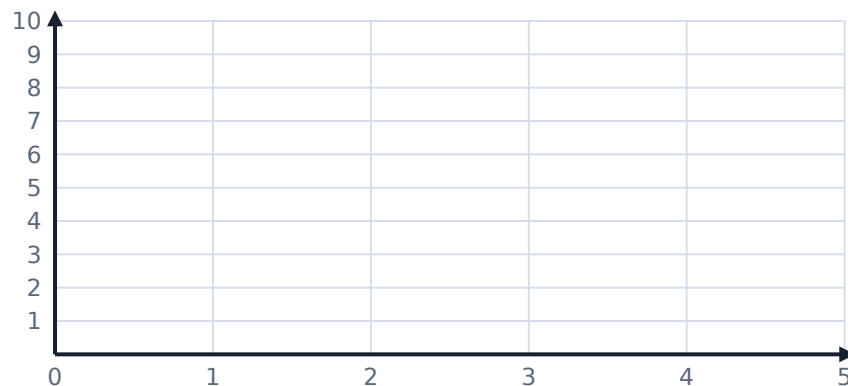
Answer: _____



Graph the pairs from the table.

Is the relationship proportional?

x	1	2	4
y	2.5	5	10



Answer: _____

True or false?

The points $(2, 1)$, $(4, 2)$, and $(6, 3)$ lie on a line through the origin, so the relationship is proportional.

- A. True
- B. False

Which set of pairs would graph as a proportional relationship?

A. (1, 1), (2, 4), (3, 9)

B. (1, 3), (2, 4), (3, 5)

C. (0, 2), (1, 4), (2, 6)

D. (2, 3), (4, 6), (6, 9)

SET 32 • 6.NS.C.6.c

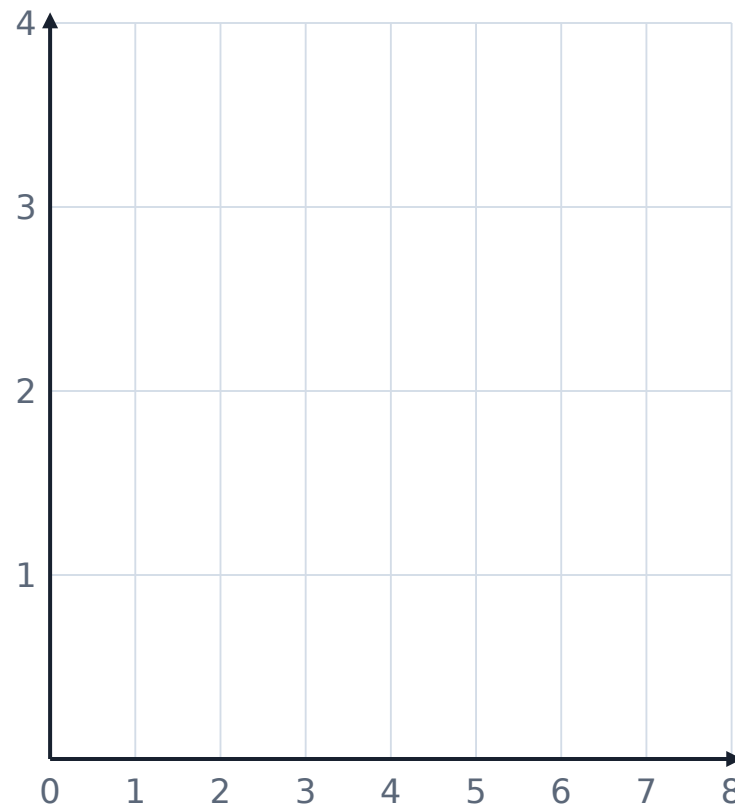
FORWARD 2

Graph proportional relationships • Grade 8 • Expressions & Equations •
8.EE.B.5

Graph $y = 0.5x$ on the coordinate plane.

What is the slope of the line?

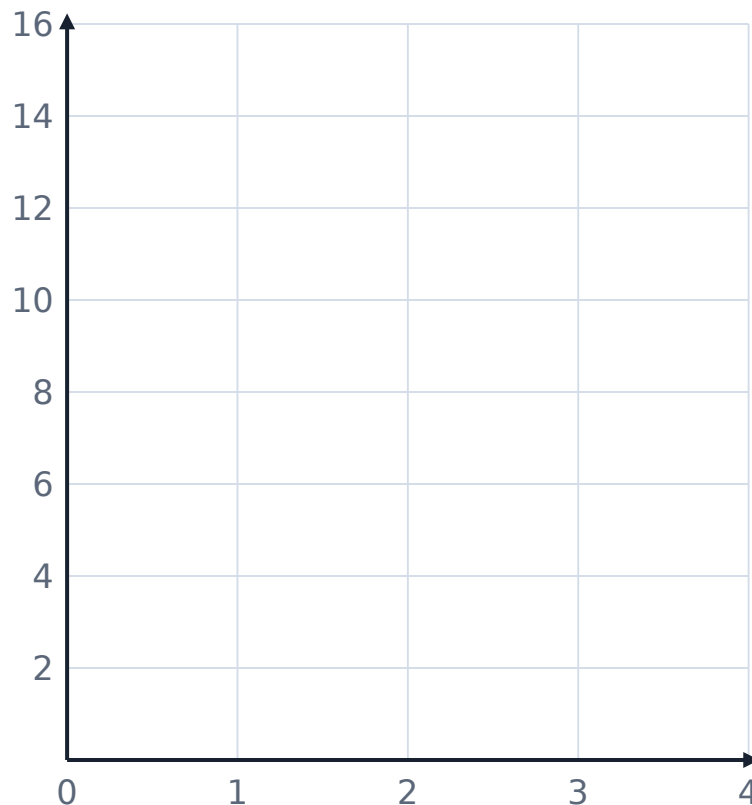
Slope: _____



Graph $y = 4x$ on the coordinate plane.

What is the slope of the line?

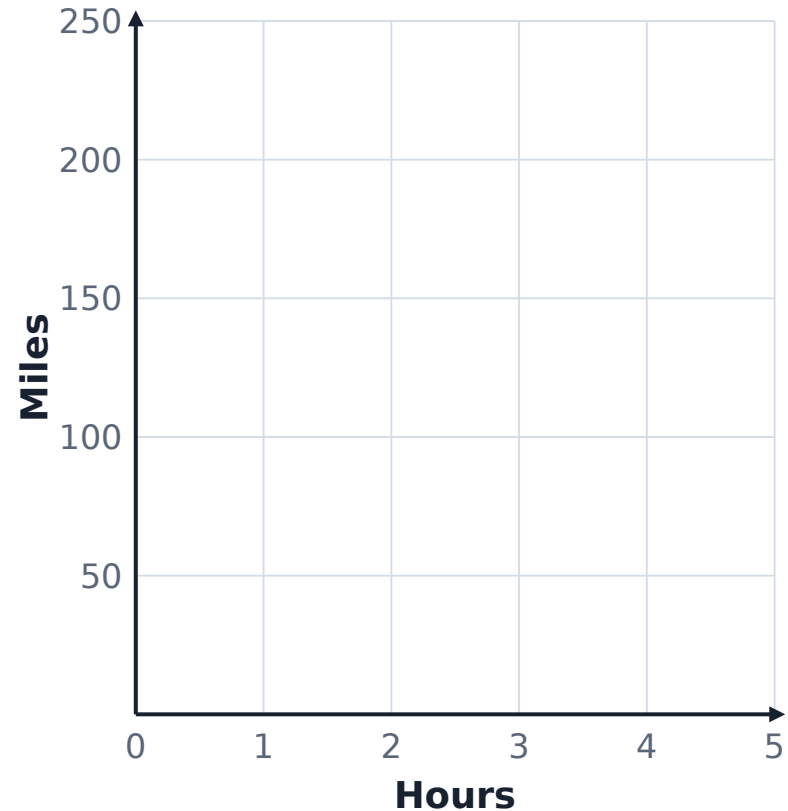
Slope: _____



A car travels 50 miles per hour. Graph the distance y after x hours.

What is the slope?

Slope: _____



Which point is on the graph of $y = 3x$?

A. (3, 1)

B. (12, 4)

C. (4, 12)

D. (4, 7)

True or false?

The graph of $y = \frac{2}{3}x$ passes through (3, 2).

A. True

B. False

Interpret inequalities as relative position on a number line

MAIN	Grade 6 main question
BACKWARD 1	Comparison symbols
BACKWARD 2	Position on a number line
BACKWARD 3	Negative numbers on a number line
FORWARD 1	Solve and graph inequalities
FORWARD 2	Compare and order irrational numbers

6.NS.C.7.a

2.NBT.A.4

2.MD.B.6

6.NS.C.6.c

7.EE.B.4.b

8.NS.A.2

SET 33 • 6.NS.C.7.a

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.a

$$-4 > -10$$

Which statement explains this inequality?

- A.** -4 is to the left of -10 on a number line.
- B.** 4 is less than 10 .
- C.** -4 is farther from 0 than -10 .
- D.** -4 is to the right of -10 on a number line.

True or false?

The number line shows that $-6 > -1$.



A. True

B. False

Write an inequality that compares -4 and -9 .

Inequality: _____

Which statement is true?

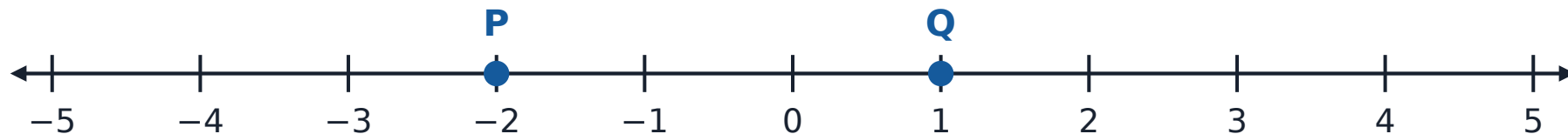
A. $-5 < -2$

B. $-2 < -5$

C. $-5 > 0$

D. $-5 > -2$

Write an inequality using the numbers at points P and Q.



Inequality: _____

SET 33 • 6.NS.C.7.a

BACKWARD 1

Comparison symbols • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Write $>$, $<$, or $=$ to compare.

452 425

Symbol: _____

True or false?

$$67 < 76$$

A. True

B. False

Which statement is true?

A. $309 < 290$

B. $309 = 290$

C. $309 > 290$

Write $>$, $<$, or $=$ to compare.

138 183

Symbol: _____

True or false?

$500 > 505$

A. True

B. False

SET 33 • 6.NS.C.7.a

BACKWARD 2

Position on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

True or false?

On a number line, 8 is to the right of 3.

A. True

B. False

Which number is farthest to the left on a number line?

A. 2

B. 5

C. 7

D. 9

Which point is farther to the left, A or B?



Point: _____

True or false?

Numbers get greater as you move to the left on a number line.

- A. True
- B. False

Which number is to the right of 14 on a number line?

- A.** 9
- B.** 11
- C.** 14
- D.** 19

SET 33 • 6.NS.C.7.a

BACKWARD 3

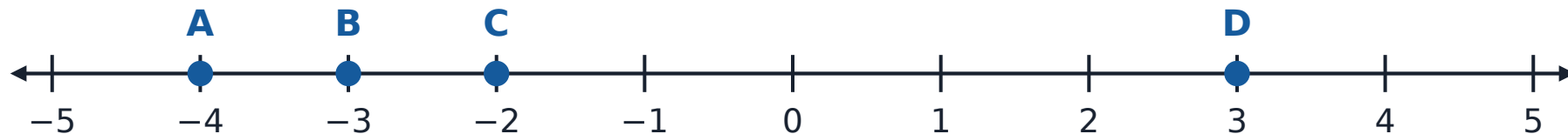
Negative numbers on a number line • Grade 6 • The Number System •
6.NS.C.6.c

What number is at point A?



A = _____

Which point is at -3 ?



A. Point A

B. Point B

C. Point C

D. Point D

True or false?

-4 is to the left of -1 on a number line.

A. True

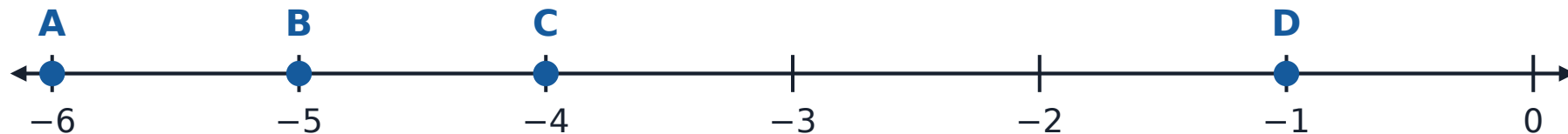
B. False

What number is at point B?



B = _____

Which point is at -5 ?



A. Point A

B. Point B

C. Point C

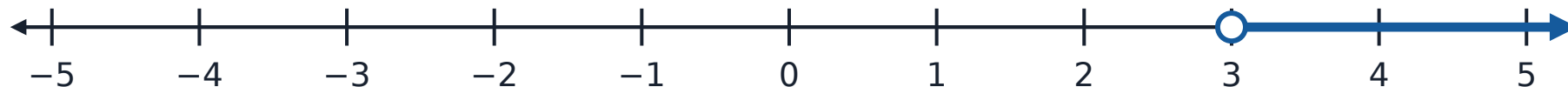
D. Point D

SET 33 • 6.NS.C.7.a

FORWARD 1

Solve and graph inequalities • Grade 7 • Expressions & Equations • 7.EE.B.4.b

Which inequality does the graph show?



A. $x > 3$

B. $x \leq 3$

C. $x < 3$

D. $x \geq 3$

Solve the inequality.

$$2x + 5 > 11$$

Solution: _____

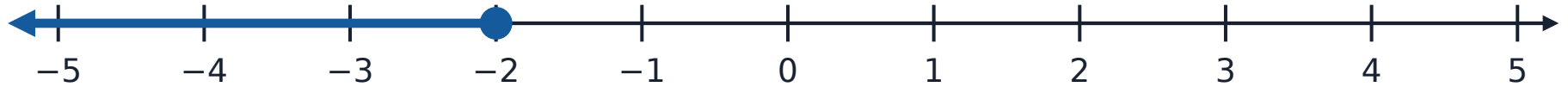
Solve the inequality.

$$-3x + 4 \leq 16$$

Solution: _____

True or false?

The graph shows the solutions of $4x - 1 \leq -9$.



A. True

B. False

Solve the inequality.

$$5x - 2 < 18$$

Solution: _____

SET 33 • 6.NS.C.7.a

FORWARD 2

Compare and order irrational numbers • Grade 8 • The Number System •
8.NS.A.2

Which is greater, $\sqrt{50}$ or 7.2?

Greater: _____

Which number is between 3 and 4?

A. $\sqrt{12}$

B. $\sqrt{8}$

C. $\sqrt{3}$

D. $\sqrt{17}$

True or false?

$$\sqrt{30} < 5.5$$

A. True

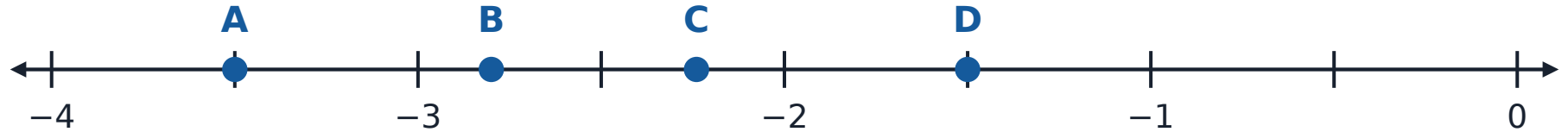
B. False

Order from least to greatest.

π , $\sqrt{8}$, 3.1

Order: _____

Which point best shows $-\sqrt{5}$?



A. Point A

B. Point B

C. Point C

D. Point D

Write, interpret, and explain order statements in real-world contexts

MAIN	Grade 6 main question	6.NS.C.7.b
BACKWARD 1	Negative numbers in context	6.NS.C.5
BACKWARD 2	Inequality symbols	1.NBT.B.3
BACKWARD 3	Order as position on a number line	6.NS.C.7.a
FORWARD 1	Write and solve inequalities for real-world situations	7.EE.B.4.b
FORWARD 2	Compare quantities written with powers of 10	8.EE.A.3

SET 34 • 6.NS.C.7.b

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.b

A temperature of -2°F is warmer than a temperature of -11°F .

Write an inequality to show this.

Inequality: _____

Account A has a balance of $-\$25$. Account B has a balance of $-\$40$.

Which statement is true?

- A.** $-25 > -40$, so Account A has the greater balance.
- B.** $-40 > -25$, so Account B has the greater balance.
- C.** $-25 < -40$, so Account A owes more.
- D.** The balances are equal.

A fish swims at -12 feet. A turtle swims at -5 feet.

Write an inequality to compare the positions. Which animal is deeper?

Inequality: _____

Deeper animal: _____

True or false?

$-15 < -10$ means that -15°F is colder than -10°F .

A. True

B. False

The lowest point in a valley is -86 meters. A beach is at 0 meters.

Which statement is true?

- A.** $-86 < 0$, so the valley's lowest point is lower.
- B.** $-86 = 0$
- C.** $0 < -86$, so the beach is lower.
- D.** $-86 > 0$, so the valley's lowest point is higher.

SET 34 • 6.NS.C.7.b

BACKWARD 1

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

Write an integer for 12 degrees below zero.

Integer: _____

True or false?

A debt of \$30 can be written as -30 .

A. True

B. False

Which situation could be represented by -6 ?

- A.** A loss of 6 yards
- B.** A deposit of \$6
- C.** A gain of 6 yards
- D.** 6 feet above sea level

Write an integer for 50 feet below sea level.

Integer: _____

True or false?

A temperature of 4 degrees above zero is written as -4 .

- A. True
- B. False

SET 34 • 6.NS.C.7.b

BACKWARD 2

Inequality symbols • Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3

Which symbol makes this true?

42 ____ 24

A. <

B. >

C. =

True or false?

$$15 > 51$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

73 37

Symbol: _____

Which symbol means "is less than"?

A. $<$

B. $>$

C. $=$

True or false?

$$89 < 98$$

A. True

B. False

SET 34 • 6.NS.C.7.b

BACKWARD 3

Order as position on a number line • Grade 6 • The Number System •
6.NS.C.7.a

True or false?

$-2 > -6$ because -2 is to the right of -6 on a number line.

A. True

B. False

Which number is greatest?

A. -9

B. -7

C. -4

D. -1

–5 is to the left of 3 on a number line.

Write an inequality to show this.

Inequality: _____

True or false?

-8 is to the right of -3 on a number line.

A. True

B. False

Which number is to the left of -4 on a number line?

A. -6

B. -2

C. 0

D. 4

SET 34 • 6.NS.C.7.b

FORWARD 1

Write and solve inequalities for real-world situations • Grade 7 • Expressions & Equations • 7.EE.B.4.b

A submarine at -40 m descends 12 m per minute. It must stay above -160 m.

Write and solve an inequality for the number of minutes t it can descend.

Solution: _____

You have \$25 and spend \$4 per day. You want to keep at least \$5.

Write and solve an inequality for the number of days d .

Solution: _____

A freezer is at 2°C . It cools 3°C per hour.

For which numbers of hours h is its temperature below -10°C ?

A. $h > 4$

B. $h > -4$

C. $h < -4$

D. $h < 4$

A diver at -8 ft descends 5 ft per minute.

Write and solve an inequality to find when the diver is deeper than -48 ft.

Solution: _____

True or false?

The solution of $-2x + 1 > 9$ is $x < -4$.

A. True

B. False

SET 34 • 6.NS.C.7.b

FORWARD 2

Compare quantities written with powers of 10 • Grade 8 • Expressions & Equations • 8.EE.A.3

A mountain is about 9×10^3 m tall. An ocean trench is about -1.1×10^4 m deep.

Which is farther from sea level?

Answer: _____

Which number is greater?

A. 3×10^{-2}

B. 8×10^{-3}

A city has about 2×10^6 people. A town has about 5×10^4 people.

How many times as many people live in the city?

Times as many: _____

Order from least to greatest.

$$4 \times 10^3, 9 \times 10^2, 1 \times 10^4$$

Order: _____

True or false?

$$-5 \times 10^3 < -2 \times 10^3$$

A. True

B. False

Find absolute value as distance from 0

MAIN	Grade 6 main question	6.NS.C.7.c
BACKWARD 1	Distance on a number line	2.MD.B.6
BACKWARD 2	Opposite numbers	6.NS.C.6.a
FORWARD 1	Distance between rational numbers as $ p - q $	7.NS.A.1.c
FORWARD 2	Square roots and cube roots as solutions	8.EE.A.2

SET 35 • 6.NS.C.7.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-18|$$

Value: _____

Which number has the greatest absolute value?

A. -12

B. -5

C. 8

D. 10

How far is -7.5 from 0 on a number line?

Distance: _____

True or false?

$$|-6| = |6|$$

A. True

B. False

Which two numbers have an absolute value of 11?

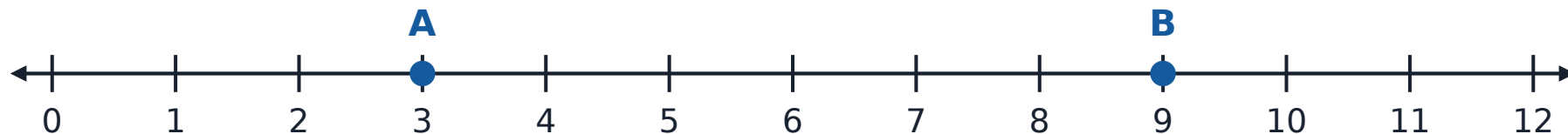
Numbers: _____

SET 35 • 6.NS.C.7.c

BACKWARD 1

Distance on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

How many units apart are points A and B?



Units: _____

True or false?

The distance from 0 to 5 is the same as the distance from 5 to 10.

- A. True
- B. False

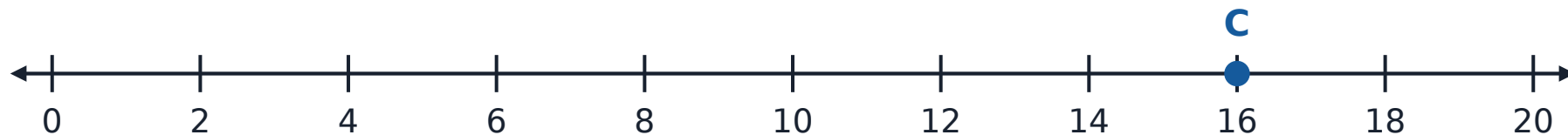
How many units from 0 is the number 11?

- A.** 0
- B.** 1
- C.** 10
- D.** 11

What is the distance from 4 to 10 on a number line?

Distance: _____

How many units is point C from 0?



Units: _____

SET 35 • 6.NS.C.7.c

BACKWARD 2

Opposite numbers • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 15?

Opposite: _____

True or false?

Opposite numbers are the same distance from 0.

A. True

B. False

Which number is the opposite of $-\frac{2}{3}$?

A. $-\frac{3}{2}$

B. $-\frac{2}{3}$

C. $\frac{2}{3}$

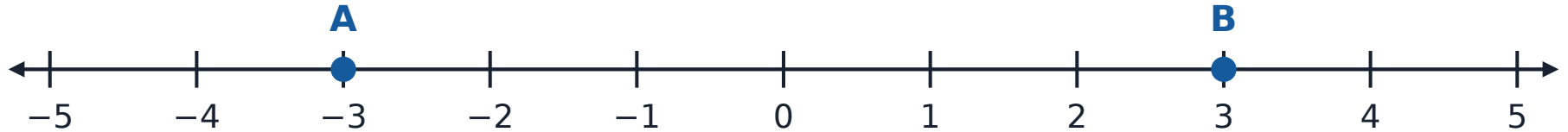
D. $\frac{3}{2}$

What is the opposite of 0?

Opposite: _____

True or false?

Points A and B represent opposite numbers.



A. True

B. False

SET 35 • 6.NS.C.7.c

FORWARD 1

Distance between rational numbers as $|p - q|$ • Grade 7 • The Number System
• 7.NS.A.1.c

Find the distance between -4 and 7 on a number line.

Distance: _____

Find the value.

$$|-3.5 - 2|$$

Value: _____

Which expression gives the distance between -9 and -2 ?

A. $-9 - 2$

B. $|-9 - (-2)|$

C. $|-9 + (-2)|$

D. $-9 + 2$

True or false?

The distance between -6 and 6 is 0 .

A. True

B. False

Find the distance between $-2\frac{1}{2}$ and $1\frac{1}{4}$ on a number line.

Distance: _____

SET 35 • 6.NS.C.7.c

FORWARD 2

Square roots and cube roots as solutions • Grade 8 • Expressions & Equations •
8.EE.A.2

Solve.

$$x^2 = 49$$

$x =$ _____

Solve.

$$x^3 = 64$$

$x =$ _____

What are the solutions of $x^2 = 81$?

- A.** $x = -9$ only
- B.** $x = 9$ only
- C.** $x = 40.5$
- D.** $x = 9$ or $x = -9$

Solve. Write the answer using a square root symbol.

$$x^2 = 20$$

$x =$ _____

True or false?

Both 6 and -6 are solutions of $x^2 = 36$.

A. True

B. False

Interpret absolute value as magnitude in a real-world situation

MAIN	Grade 6 main question	6.NS.C.7.c
BACKWARD 1	Negative numbers in context	6.NS.C.5
BACKWARD 2	Distance from 0	2.MD.B.6
BACKWARD 3	Opposites are the same distance from 0	6.NS.C.6.a
FORWARD 1	Real-world problems with rational numbers	7.NS.A.3
FORWARD 2	Compare magnitudes with powers of 10	8.EE.A.3

SET 36 • 6.NS.C.7.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.c

An account balance is $-\$45$.

What is the absolute value of the balance, and what does it describe?

Absolute value: _____

It describes: _____

A diver is at an elevation of -42 feet.

How far is the diver from sea level?

Distance: _____

A stock's value changed by $-\$8.50$ today.

What is the size of the change?

A. $-\$8.50$

B. $\$0$

C. $\$8.50$

D. $\$17.00$

The temperature is -13°F .

How many degrees below zero is it?

Degrees: _____

A submarine is at -275 m. A plane is at 275 m.

Which is farther from sea level? Explain using absolute value.

Answer: _____

SET 36 • 6.NS.C.7.c

BACKWARD 1

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

True or false?

–30 can represent owing \$30.

A. True

B. False

Write an integer for 42 feet below sea level.

Integer: _____

Which situation is represented by $+25$?

- A.** A loss of 25 yards
- B.** 25 feet below sea level
- C.** A debt of \$25
- D.** A deposit of \$25

Write an integer for a debt of \$18.

Integer: _____

True or false?

A temperature of 0°F means there is no temperature.

A. True

B. False

SET 36 • 6.NS.C.7.c

BACKWARD 2

Distance from 0 • Grade 2 • Measurement & Data • 2.MD.B.6

How many units from 0 is point A?



Units: _____

True or false?

The distance from 0 to 13 is 13 units.

A. True

B. False

Which number is 5 units from 0 on this number line?



A. 0

B. 5

C. 10

D. 50

How far is 20 from 0 on a number line?

Distance: _____

True or false?

The number that is 0 units from 0 is 1.

A. True

B. False

SET 36 • 6.NS.C.7.c

BACKWARD 3

Opposites are the same distance from 0 • Grade 6 • The Number System •
6.NS.C.6.a

True or false?

–42 and 42 are the same distance from 0.

A. True

B. False

What is the opposite of -275 ?

Opposite: _____

Which number is the same distance from 0 as -8.5 ?

A. -5.8

B. 0

C. 8.5

D. 85

True or false?

The opposite of 13 is -13 .

A. True

B. False

Name the number that is the same distance from 0 as -45 , on the other side of 0.

Number: _____

SET 36 • 6.NS.C.7.c

FORWARD 1

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A company lost \$2.5 million in one year and earned a profit of \$1.8 million the next year.

What is the total for the two years?

Total: _____

The temperature was -6°F . It dropped 11°F .
What is the new temperature?

Temperature: _____

A diver at -35.5 feet rises 12.25 feet.

What is the diver's new position?

- A.** -47.75 feet
- B.** -23.25 feet
- C.** -22.75 feet
- D.** 23.25 feet

A hiker goes from an elevation of 1,200 feet to an elevation of -80 feet.

What is the change in elevation?

Change: _____

True or false?

If the temperature drops from -4°C to -9°C , it gets 5°C colder.

A. True

B. False

SET 36 • 6.NS.C.7.c

FORWARD 2

Compare magnitudes with powers of 10 • Grade 8 • Expressions & Equations •
8.EE.A.3

One company's debt is 2×10^6 dollars. Another company's debt is 4×10^4 dollars.

How many times as large is the first debt?

Times as large: _____

The deepest ocean point is about -1×10^4 meters. A lake bottom is about -5×10^1 meters.

About how many times as deep is the ocean point?

A. 5

B. 20

C. 200

D. 2,000

Estimate 7,980,000 as a single digit times a power of 10.

Estimate: _____

True or false?

6×10^9 is 300 times as large as 2×10^7 .

A. True

B. False

Estimate 0.0000031 as a single digit times a power of 10.

Estimate: _____

Distinguish comparisons of absolute value from statements about order

MAIN	Grade 6 main question	6.NS.C.7.d
BACKWARD 1	Absolute value	6.NS.C.7.c
BACKWARD 2	Ordering negative numbers	6.NS.C.7.a
BACKWARD 3	Negative numbers in context	6.NS.C.5
FORWARD 1	Real-world problems with rational numbers	7.NS.A.3
FORWARD 2	Compare the size and order of irrational numbers	8.NS.A.2

SET 37 • 6.NS.C.7.d

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.d

True or false?

An account balance less than $-\$50$ means a debt greater than $\$50$.

- A. True
- B. False

Which account balance shows the greatest debt?

A. $-\$55$

B. $-\$38$

C. $-\$20$

D. $\$10$

True or false?

$-12 < -5$, so $|-12| < |-5|$.

A. True

B. False

Two temperatures are -15°F and -9°F .

Which temperature is colder? Which has the greater absolute value?

Colder: _____

Greater absolute value: _____

A submarine is at -300 feet. A diver is at -45 feet.

Which statement is true?

- A.** The diver is deeper because $-45 > -300$.
- B.** The submarine is deeper because $-300 > -45$.
- C.** The submarine is deeper because $|-300| > |-45|$.
- D.** They are at the same depth.

SET 37 • 6.NS.C.7.d

BACKWARD 1

Absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-25|$$

Value: _____

Find the value.

$|14|$

Value: _____

True or false?

$$|-3| = -3$$

A. True

B. False

Which number has an absolute value of 7?

A. -7

B. $-\frac{1}{7}$

C. 0

D. $\frac{1}{7}$

Find the value.

$$|-2.5|$$

Value: _____

SET 37 • 6.NS.C.7.d

BACKWARD 2

Ordering negative numbers • Grade 6 • The Number System • 6.NS.C.7.a

Which number is least?

A. -20

B. -12

C. -5

D. 0

True or false?

$$-9 > -4$$

A. True

B. False

Order from least to greatest.

$-3, -8, 1$

Order: _____

True or false?

$$-1 > -100$$

A. True

B. False

Which number is greater?

A. -16

B. -6

SET 37 • 6.NS.C.7.d

BACKWARD 3

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

Write an integer for a debt of \$40.

Integer: _____

What could -8 represent?

- A.** A deposit of \$8
- B.** A gain of 8 yards
- C.** 8 feet below sea level
- D.** 8 feet above sea level

True or false?

An account balance of $-\$10$ means you owe $\$10$.

- A. True
- B. False

Write an integer for 15 degrees below zero.

Integer: _____

True or false?

A 20-foot descent can be written as +20.

A. True

B. False

SET 37 • 6.NS.C.7.d

FORWARD 1

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

Two accounts have balances of $-\$42.50$ and $-\$18.75$.
How much greater is the larger debt?

Difference: _____

A diver at -24.5 m rises 9.75 m and then descends 15 m.

What is the diver's final position?

Position: _____

The temperature at 6 a.m. was -8.5°F . At noon it was 3.5°F .

How much did the temperature rise?

A. -12°F

B. -5°F

C. 5°F

D. 12°F

Maria's balance is $-\$36$. She pays back $\frac{1}{4}$ of what she owes.

What is her new balance?

Balance: _____

True or false?

A change from -12 m to -30 m is a change of -18 m.

A. True

B. False

SET 37 • 6.NS.C.7.d

FORWARD 2

Compare the size and order of irrational numbers • Grade 8 • The Number System • 8.NS.A.2

Which number has the greater absolute value?

A. 4.3

B. $-\sqrt{20}$

True or false?

$$-\sqrt{10} < -3$$

A. True

B. False

Order from least to greatest.

$-\sqrt{5}$, -2.5 , -2

Order: _____

Which number is farther from 0 on a number line?

A. 3.1

B. $-\pi$

Between which two consecutive integers is $-\sqrt{30}$?

Between: _____

Find distances between points with the same first or second coordinate

MAIN	Grade 6 main question	6.NS.C.8
BACKWARD 1	Absolute value	6.NS.C.7.c
BACKWARD 2	Naming points on the coordinate plane	6.NS.C.6.c
BACKWARD 3	Distance between whole numbers	2.MD.B.6
FORWARD 1	Distance between rational numbers	7.NS.A.1.c
FORWARD 2	Distance between points using the Pythagorean Theorem	8.G.B.8

SET 38 • 6.NS.C.8

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(-3, 4)$ and $(5, 4)$.

Distance: _____

Find the distance between $(2, -6)$ and $(2, 5)$.

Distance: _____

What is the distance between $(-7, -2)$ and $(-1, -2)$?

A. -6

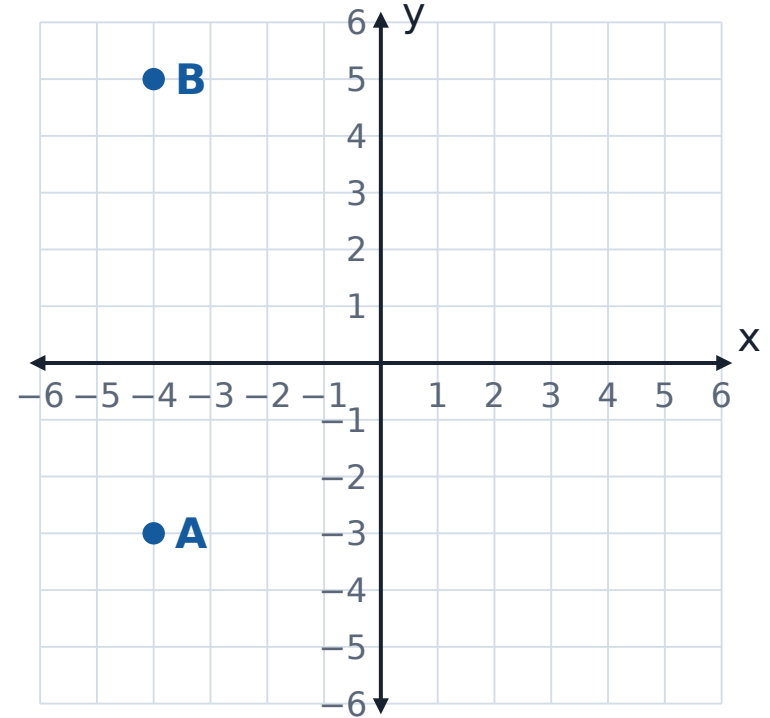
B. 3

C. 6

D. 8

What is the distance between points A and B?

Distance: _____



Find the distance between $(-2.5, 1)$ and $(3.5, 1)$.

Distance: _____

SET 38 • 6.NS.C.8

BACKWARD 1

Absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-7|$$

Value: _____

True or false?

$$|-3| + |5| = 8$$

A. True

B. False

$$|-4| + |6| = ?$$

A. -10

B. -2

C. 2

D. 10

Find the value.

$$|-9| + |2|$$

Value: _____

True or false?

The distance from -5 to 0 is $|-5|$.

A. True

B. False

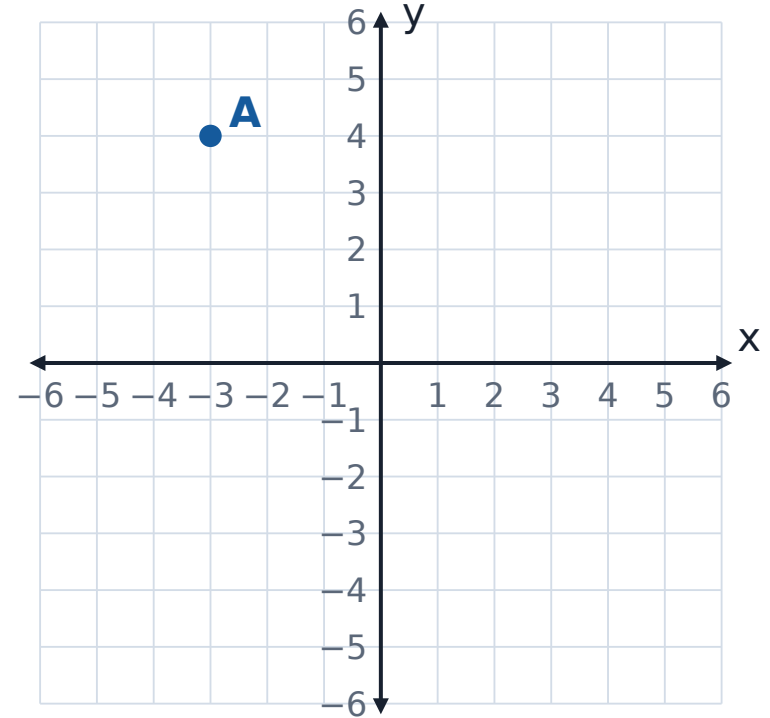
SET 38 • 6.NS.C.8

BACKWARD 2

Naming points on the coordinate plane • Grade 6 • The Number System •
6.NS.C.6.c

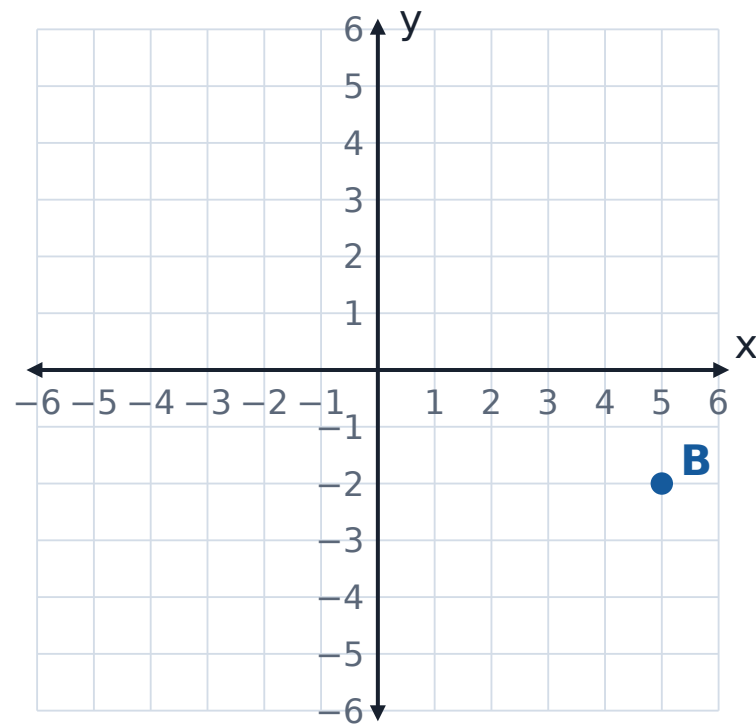
What are the coordinates of point A?

A = _____



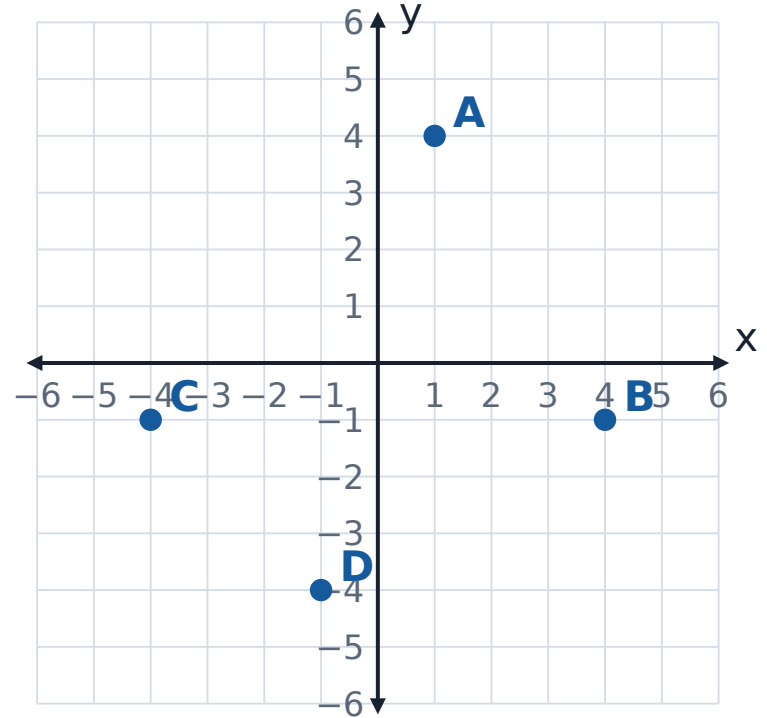
What are the coordinates of point B?

B = _____



Which point is at $(-1, -4)$?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D

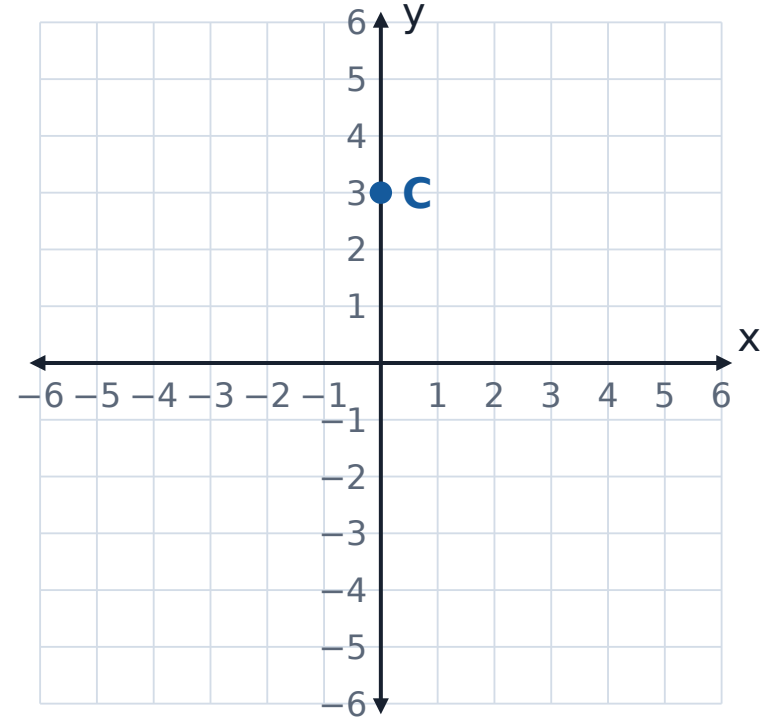


True or false?

Point C is located at $(3, 0)$.

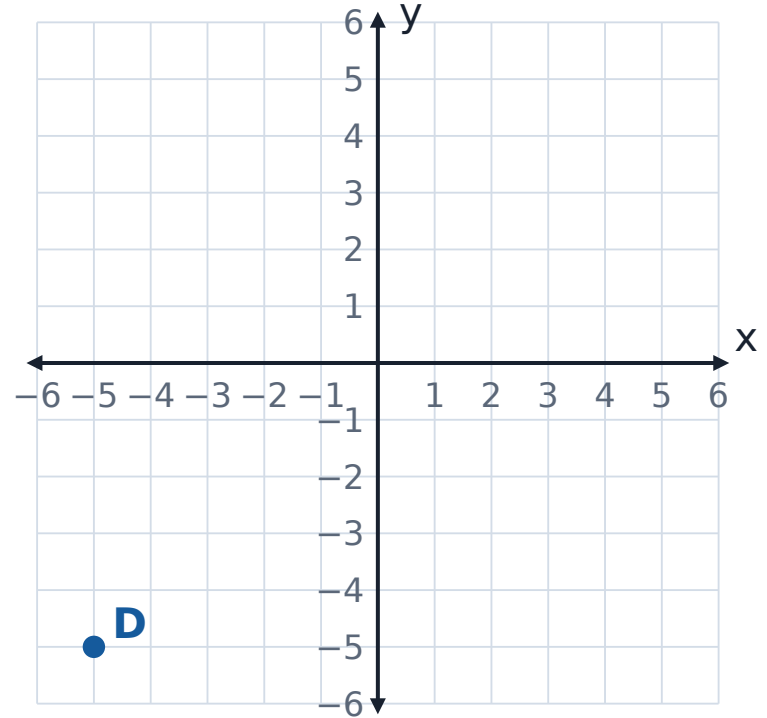
A. True

B. False



What are the coordinates of point D?

D = _____



SET 38 • 6.NS.C.8

BACKWARD 3

Distance between whole numbers • Grade 2 • Measurement & Data • 2.MD.B.6

What is the distance from 3 to 11 on a number line?

Distance: _____

How many units apart are points A and B?



Units: _____

True or false?

The distance from 12 to 4 is 8 units.

A. True

B. False

What is the distance from 5 to 14?

- A.** 8
- B.** 9
- C.** 10
- D.** 19

What is the distance from 0 to 13 on a number line?

Distance: _____

SET 38 • 6.NS.C.8

FORWARD 1

Distance between rational numbers • Grade 7 • The Number System •
7.NS.A.1.c

Find the distance between -2.5 and 4.5 on a number line.

Distance: _____

Find the value.

$$|-8.4 - 3.6|$$

Value: _____

What is the distance between $-3\frac{1}{2}$ and $2\frac{1}{4}$ on a number line?

A. $-5\frac{3}{4}$

B. $1\frac{1}{4}$

C. $5\frac{3}{4}$

D. $6\frac{1}{4}$

True or false?

The distance between -7 and -2 is $|-7 - (-2)|$, which is 5.

A. True

B. False

Find the distance between -4 and 2.5 on a number line.

Distance: _____

SET 38 • 6.NS.C.8

FORWARD 2

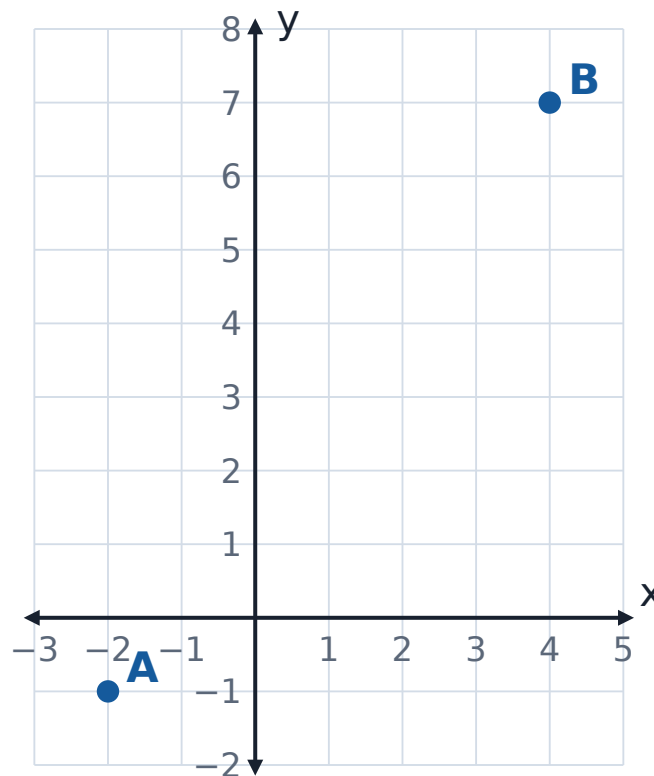
Distance between points using the Pythagorean Theorem • Grade 8 •
Geometry • 8.G.B.8

Find the distance between $(1, 2)$ and $(4, 6)$.

Distance: _____

Find the distance between points A and B.

Distance: _____



What is the distance between (0, 0) and (5, 12)?

A. $\sqrt{119}$

B. 17

C. 13

D. 7

Find the distance between $(-3, 5)$ and $(1, 2)$.

Distance: _____

True or false?

The distance between $(2, 1)$ and $(5, 5)$ is 5.

A. True

B. False

Solve real-world problems by graphing points in all four quadrants

MAIN	Grade 6 main question	6.NS.C.8
BACKWARD 1	Absolute value as distance	6.NS.C.7.c
BACKWARD 2	Plotting points in all four quadrants	6.NS.C.6.c
BACKWARD 3	Perimeter of rectangles	3.MD.D.8
BACKWARD 4	Area of rectangles	3.MD.C.7.b
FORWARD 1	Use a map scale on a coordinate grid	7.G.A.1
FORWARD 2	Straight-line distances on a map	8.G.B.8

SET 39 • 6.NS.C.8

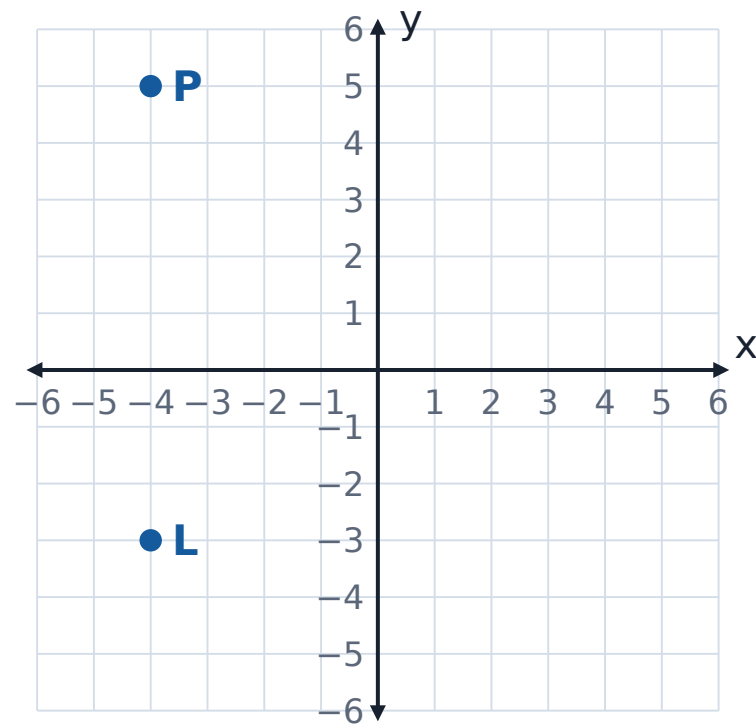
MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.8

On the map, each unit is 1 block.

How many blocks is it from the library L to the park P?

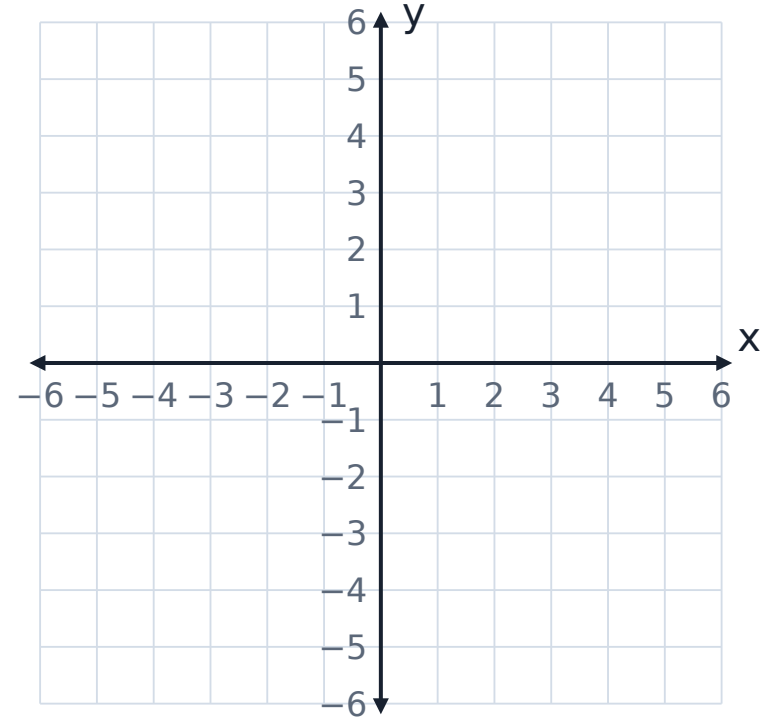
Blocks: _____



A park is a rectangle with corners at $(-2, 3)$, $(4, 3)$, $(4, -1)$, and $(-2, -1)$. Each unit is 1 meter.

Graph the park. What is its perimeter?

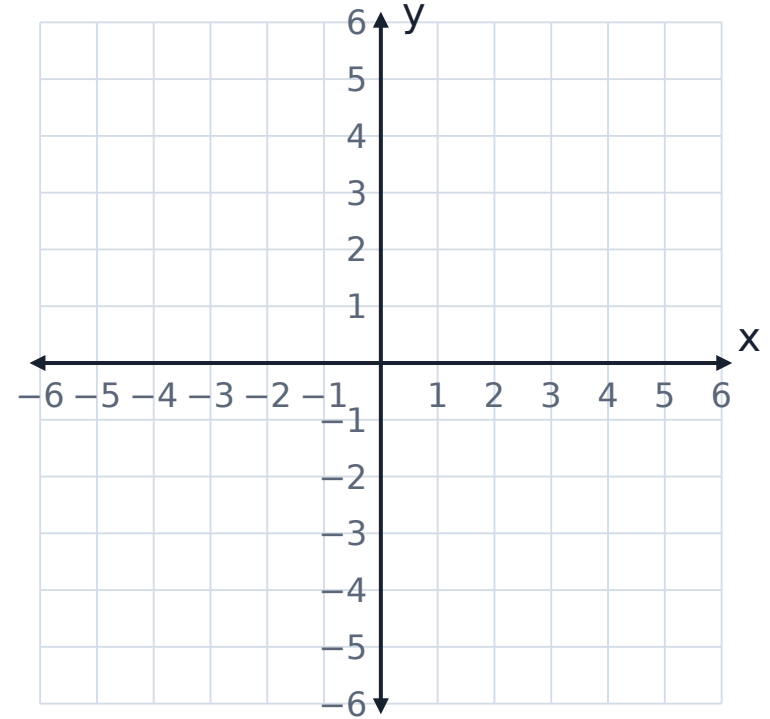
Perimeter: _____



The school is at $(-3, 2)$ and the pool is at $(-3, -4)$. Each unit is 1 block.

Graph both places. How many blocks apart are they?

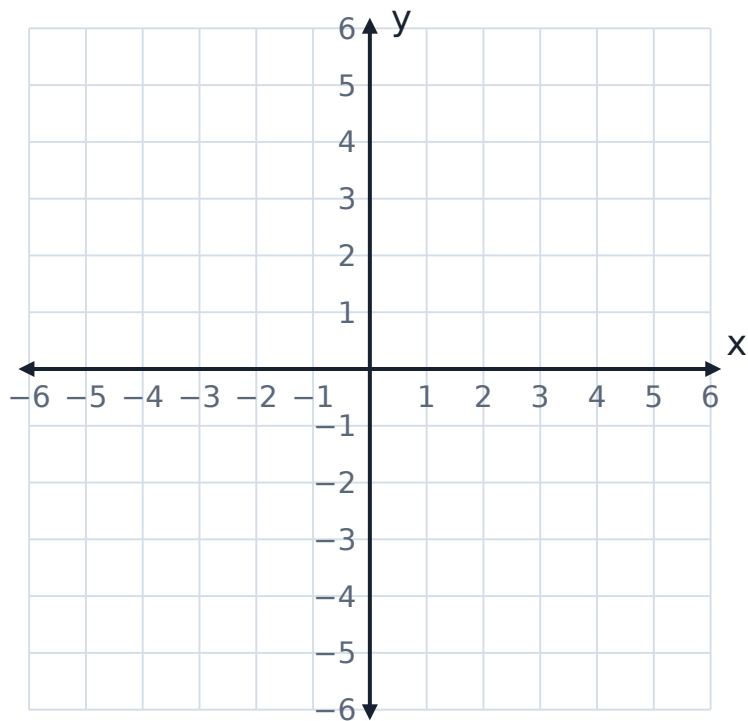
Blocks: _____



A garden is a rectangle with corners at $(-5, 1)$, $(1, 1)$, $(1, -3)$, and $(-5, -3)$. Each unit is 1 meter.

What is the area of the garden?

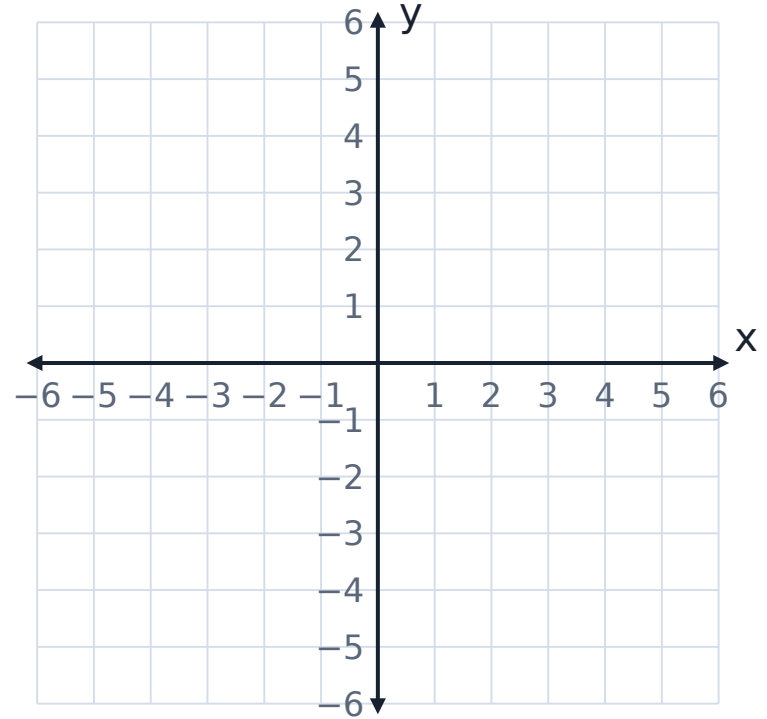
- | | |
|----------------------------|----------------------------|
| A. 10 square meters | B. 12 square meters |
| C. 20 square meters | D. 24 square meters |



A hiker starts at $(-5, 0)$, walks to $(3, 0)$, and then walks to $(3, 6)$. Each unit is 1 km.

Graph the path. How far does the hiker walk?

Distance: _____



SET 39 • 6.NS.C.8

BACKWARD 1

Absolute value as distance • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-4| + |5|$$

Value: _____

True or false?

The distance from -3 to 0 is 3 .

A. True

B. False

How far is -6 from 0 ?

A. -6

B. 0

C. 6

D. 12

Find the value.

$$|-2| + |4|$$

Value: _____

True or false?

$$|-5| = -5$$

A. True

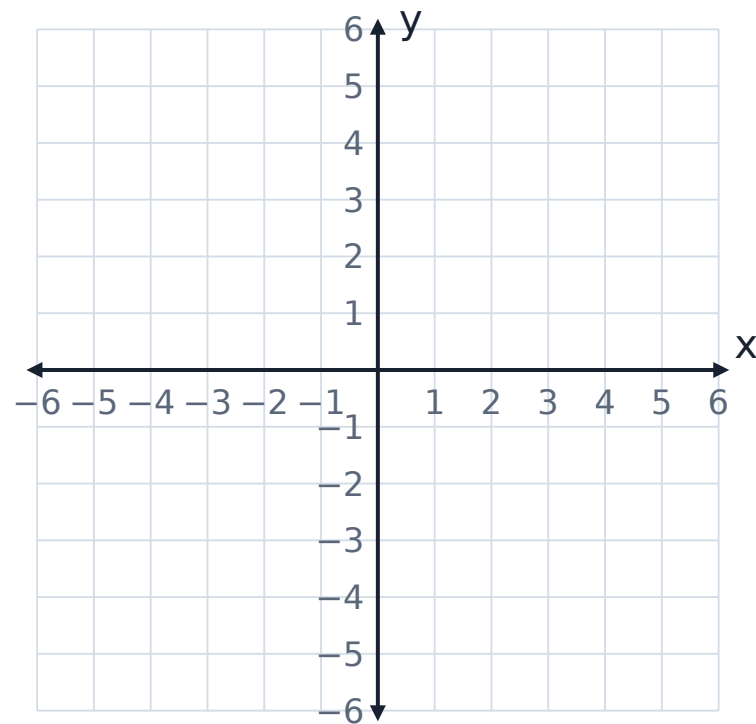
B. False

SET 39 • 6.NS.C.8

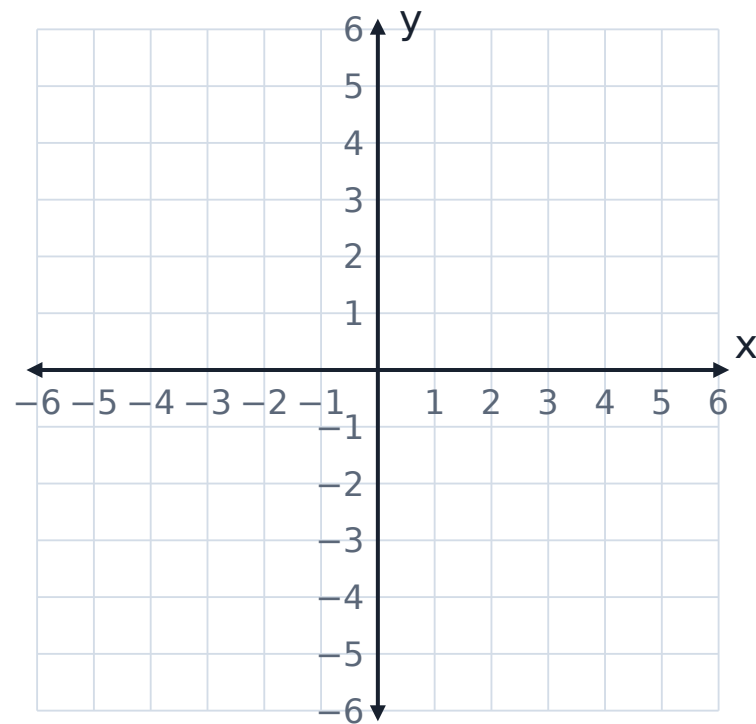
BACKWARD 2

Plotting points in all four quadrants • Grade 6 • The Number System •
6.NS.C.6.c

Plot and label point A($-2, 3$).



Plot and label point B(4, -1).



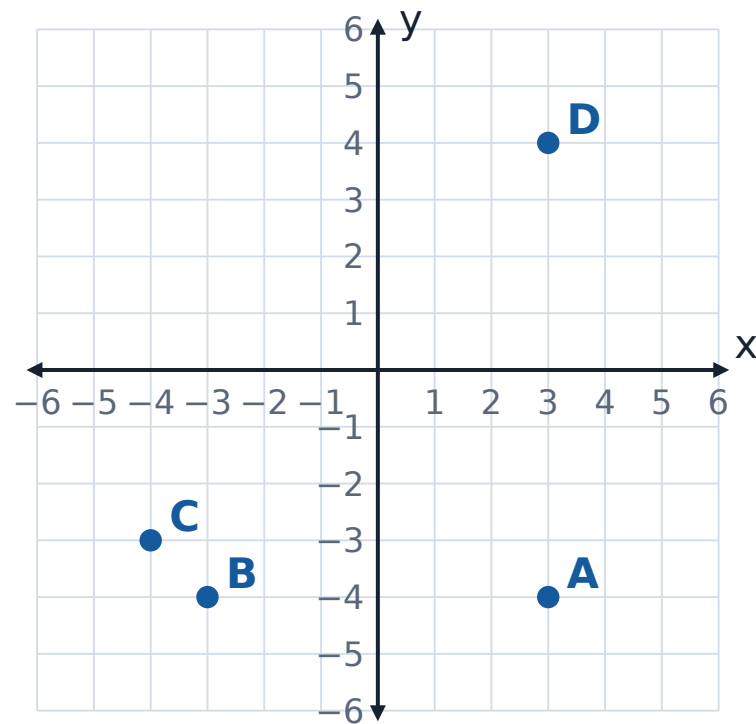
Which point is at $(-3, -4)$?

A. Point A

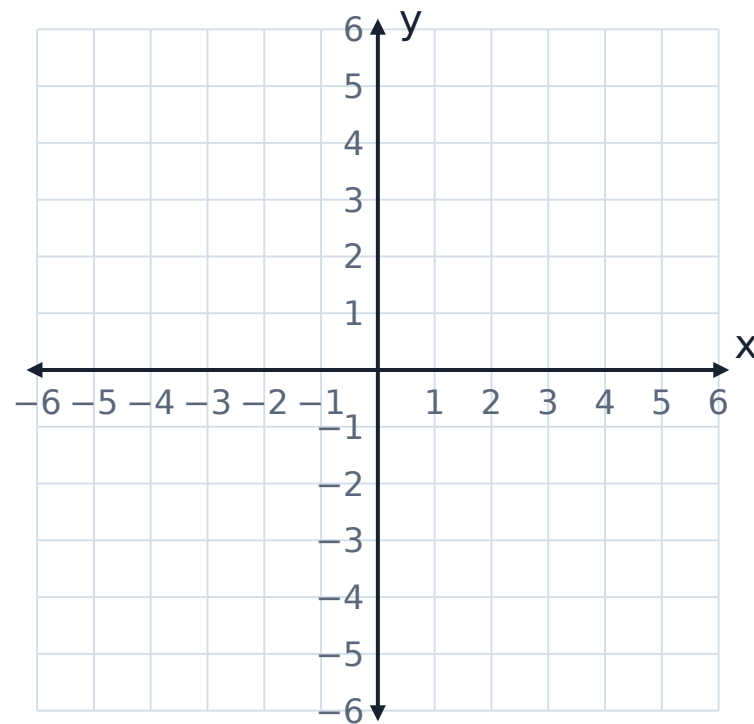
B. Point B

C. Point C

D. Point D



Plot and label point C(0, -5).

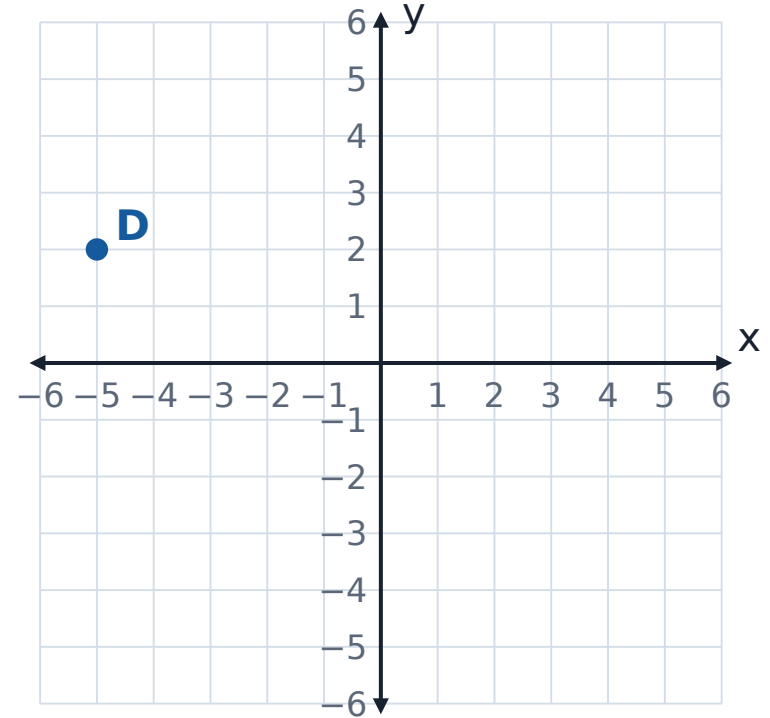


True or false?

Point D is located at $(-5, 2)$.

A. True

B. False



SET 39 • 6.NS.C.8

BACKWARD 3

Perimeter of rectangles • Grade 3 • Measurement & Data • 3.MD.D.8

What is the perimeter of the rectangle?

Perimeter: _____



A rectangle is 9 cm long and 3 cm wide.

What is its perimeter?

Perimeter: _____

True or false?

A square with 5-meter sides has a perimeter of 25 meters.

A. True

B. False

A rectangle is 8 inches long and 2 inches wide.

What is its perimeter?

- A.** 10 inches
- B.** 16 inches
- C.** 18 inches
- D.** 20 inches

A rectangle has a perimeter of 18 feet and a length of 5 feet.

What is its width?

Width: _____

SET 39 • 6.NS.C.8

BACKWARD 4

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rectangle is 6 m long and 4 m wide.

What is its area?

Area: _____

True or false?

A rectangle that is 7 units by 3 units has an area of 21 square units.

- A.** True
- B.** False

A rectangle is 5 ft by 8 ft.

What is its area?

- A.** 13 square feet
- B.** 26 square feet
- C.** 40 square feet
- D.** 80 square feet

What is the area of the rectangle?

Area: _____



A square has 6-inch sides.

What is its area?

Area: _____

SET 39 • 6.NS.C.8

FORWARD 1

Use a map scale on a coordinate grid • Grade 7 • Geometry • 7.G.A.1

On a map grid, each unit is $\frac{1}{2}$ mile. The school is at $(-3, 2)$ and the park is at $(5, 2)$.

What is the actual distance between them?

Distance: _____

On a trail map, each unit is 250 meters. The trail goes straight from $(0, -4)$ to $(0, 6)$.

How long is the actual trail?

Length: _____

On a map, each unit is 1.5 km. A park has corners at $(-2, 1)$, $(4, 1)$, $(4, -3)$, and $(-2, -3)$.

What is the actual perimeter of the park?

A. 13.3 km

B. 20 km

C. 30 km

D. 36 km

On a plan, each unit is 20 feet. A garden has corners at $(-1, -1)$, $(3, -1)$, $(3, 2)$, and $(-1, 2)$.

What are the actual length and width?

Length and width: _____

True or false?

On a map where each unit is 5 miles, the points $(-3, 0)$ and $(3, 0)$ are 30 miles apart.

A. True

B. False

SET 39 • 6.NS.C.8

FORWARD 2

Straight-line distances on a map • Grade 8 • Geometry • 8.G.B.8

On a map, the library is at $(-2, 1)$ and the pool is at $(4, 9)$. Each unit is 1 block.

What is the straight-line distance between them?

Distance: _____

A drone flies from $(0, 0)$ to $(9, 12)$. Each unit is 1 meter.
How far does it fly in a straight line?

Distance: _____

Home is at $(-3, -1)$ and school is at $(1, 2)$. Each unit is 1 km.

What is the straight-line distance?

A. 5 km

B. 25 km

C. 7 km

D. $\sqrt{7}$ km

Find the straight-line distance from $(-4, 6)$ to $(2, -2)$.

Distance: _____

True or false?

The straight-line distance from $(1, 1)$ to $(7, 9)$ is 10.

A. True

B. False

Evaluate numerical expressions with whole-number exponents

MAIN	Grade 6 main question	6.EE.A.1
BACKWARD 1	Powers of 10 with exponents	5.NBT.A.2
BACKWARD 2	Order of operations with parentheses	5.OA.A.1
BACKWARD 3	Repeated multiplication	3.OA.C.7
FORWARD 1	Products of signed numbers	7.NS.A.2.a
FORWARD 2	Evaluate expressions with integer exponents	8.EE.A.1

SET 40 • 6.EE.A.1

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$2^4 + 3^2$$

Value: _____

Evaluate.

$$3 \times 2^3$$

Value: _____

Evaluate.

$$(5 - 2)^3 \div 9$$

Value: _____

Evaluate.

$$5^2 - 4 \times 2$$

A. -3

B. 17

C. 21

D. 42

Evaluate.

$$\left(\frac{2}{3}\right)^2$$

Value: _____

SET 40 • 6.EE.A.1

BACKWARD 1

Powers of 10 with exponents • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Write 10^3 as a whole number.

Number: _____

True or false?

$$10^4 = 10,000$$

A. True

B. False

Which is equal to 100,000?

A. 10^5

B. 5^{10}

C. 10^4

D. 10^6

Write 1,000,000 as a power of 10.

Power of 10: _____

True or false?

$$10^2 = 20$$

A. True

B. False

SET 40 • 6.EE.A.1

BACKWARD 2

Order of operations with parentheses • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$(8 + 4) \div 3$$

Value: _____

Evaluate.

$$20 - (3 \times 4)$$

Value: _____

True or false?

$$2 \times (3 + 5) = 11$$

A. True

B. False

Evaluate.

$$18 \div (2 + 4)$$

A. 3

B. 6

C. 9

D. 13

Evaluate.

$$(9 - 3) \times 5$$

Value: _____

SET 40 • 6.EE.A.1

BACKWARD 3

Repeated multiplication • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$2 \times 2 \times 2$$

Product: _____

Multiply.

$$5 \times 5$$

Product: _____

True or false?

$$3 \times 3 \times 3 = 9$$

A. True

B. False

$$4 \times 4 \times 4 = ?$$

A. 12

B. 16

C. 44

D. 64

Multiply.

$$6 \times 6$$

Product: _____

SET 40 • 6.EE.A.1

FORWARD 1

Products of signed numbers • Grade 7 • The Number System • 7.NS.A.2.a

Evaluate.

$$(-3)^2$$

Value: _____

Evaluate.

$$(-2)^3$$

Value: _____

True or false?

$$-3^2 = (-3)^2$$

A. True

B. False

$$(-1)^5 = ?$$

A. -5

B. -1

C. 1

D. 5

Evaluate.

$$\left(-\frac{1}{2}\right)^2$$

Value: _____

SET 40 • 6.EE.A.1

FORWARD 2

Evaluate expressions with integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Evaluate.

$$2^{-3}$$

Value: _____

Evaluate.

$$5^{-2} \times 5^4$$

Value: _____

$$10^{-2} = ?$$

A. -100

B. -20

C. $\frac{1}{100}$

D. 0.1

Evaluate.

$$\frac{4^5}{4^3}$$

Value: _____

True or false?

$$7^0 = 1$$

A. True

B. False

Write numerical expressions with whole-number exponents

MAIN	Grade 6 main question	6.EE.A.1
BACKWARD 1	Powers of 10 with exponents	5.NBT.A.2
BACKWARD 2	Multiplying with repeated factors	3.OA.C.7
FORWARD 1	Powers of negative numbers	7.NS.A.2.a
FORWARD 2	Write expressions as a single power	8.EE.A.1

SET 41 • 6.EE.A.1

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.1

Write using an exponent.

$$7 \times 7 \times 7 \times 7$$

Expression: _____

Which expression is equal to $3 \times 3 \times 3 \times 3 \times 3$?

A. 5^3

B. 15

C. 3^5

D. 3×5

Write using an exponent.

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

Expression: _____

Write using an exponent.

$$0.4 \times 0.4$$

Expression: _____

Write using exponents.

$$2 \times 2 \times 2 \times 5 \times 5$$

Expression: _____

SET 41 • 6.EE.A.1

BACKWARD 1

Powers of 10 with exponents • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Write $10 \times 10 \times 10$ as a power of 10.

Power of 10: _____

True or false?

$$10 \times 10 \times 10 \times 10 = 10^4$$

A. True

B. False

Which is equal to 10^5 ?

A. $5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5$

B. $10 \times 10 \times 10 \times 10 \times 10$

C. 100,000,000

D. 10×5

How many factors of 10 are in 10^6 ?

Factors: _____

True or false?

$$10^2 = 10 \times 2$$

A. True

B. False

SET 41 • 6.EE.A.1

BACKWARD 2

Multiplying with repeated factors • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$2 \times 2 \times 2 \times 2$$

Product: _____

Multiply.

$$3 \times 3$$

Product: _____

True or false?

$$5 \times 5 \times 5 = 15$$

A. True

B. False

How many times is 7 used as a factor in $7 \times 7 \times 7$?

- A.** 1
- B.** 3
- C.** 7
- D.** 21

Multiply.

$$4 \times 4$$

Product: _____

SET 41 • 6.EE.A.1

FORWARD 1

Powers of negative numbers • Grade 7 • The Number System • 7.NS.A.2.a

Write using an exponent.

$$(-3)(-3)(-3)(-3)$$

Expression: _____

Write $(-2)(-2)(-2)$ using an exponent. Is its value positive or negative?

Expression: _____

Sign: _____

Which expression is equal to $(-5)^2$?

A. -5×2

B. $(-5)(-5)$

C. -5×5

D. $-(5 \times 5)$

True or false?

$(-1)^6$ is positive.

A. True

B. False

Write using an exponent.

$$\left(-\frac{2}{3}\right)\left(-\frac{2}{3}\right)$$

Expression: _____

SET 41 • 6.EE.A.1

FORWARD 2

Write expressions as a single power • Grade 8 • Expressions & Equations •
8.EE.A.1

Write as a single power.

$$4^3 \times 4^5$$

Single power: _____

Write as a single power.

$$(2^3)^4$$

Single power: _____

Which is equal to $\frac{6^7}{6^2}$?

A. 6^{14}

B. 6^5

C. 6^9

D. 1^5

Write as a single power.

$$3^4 \times 3^{-6}$$

Single power: _____

True or false?

$$5^2 \times 5^3 = 5^6$$

A. True

B. False

Write expressions in which letters stand for numbers

MAIN	Grade 6 main question	6.EE.A.2.a
BACKWARD 1	Words for operations	5.OA.A.2
BACKWARD 2	A symbol for an unknown number	3.OA.A.4
BACKWARD 3	Order matters in subtraction and division	3.OA.B.5
FORWARD 1	Write and solve equations of the form $px + q = r$	7.EE.B.4.a
FORWARD 2	Construct a function from a description	8.F.B.4

SET 42 • 6.EE.A.2.a

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.a

Write an expression for "subtract m from 12."

Expression: _____

Write an expression for "5 more than twice a number n ."

Expression: _____

Which expression means "the quotient of x and 4"?

A. $4x$

B. $\frac{4}{x}$

C. $x - 4$

D. $\frac{x}{4}$

Write an expression for "3 less than the product of 7 and a number m ."

Expression: _____

True or false?

"8 less than k" can be written as $8 - k$.

A. True

B. False

SET 42 • 6.EE.A.2.a

BACKWARD 1

Words for operations • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Which expression shows "add 6 and 9, then multiply by 4"?

A. $6 + 9 \times 4$

B. $4 \times (6 + 9)$

C. $(4 + 6) \times 9$

D. $4 \times 6 + 9$

Write an expression for "subtract 4 from 10, then divide by 2."

Expression: _____

True or false?

"The product of 6 and 9" means 6×9 .

A. True

B. False

Which word describes the answer to a subtraction problem?

- A.** Quotient
- B.** Difference
- C.** Product
- D.** Sum

Write an expression for "the sum of 12 and 5, multiplied by 3."

Expression: _____

SET 42 • 6.EE.A.2.a

BACKWARD 2

A symbol for an unknown number • Grade 3 • Operations & Algebraic Thinking
• 3.OA.A.4

What number goes in the box?

$$\square + 7 = 12$$

$$\square = \underline{\hspace{10em}}$$

True or false?

In $7 \times \square = 42$, the box stands for 6.

A. True

B. False

$$\square - 5 = 9$$

What number goes in the box?

- A.** 4
- B.** 9
- C.** 14
- D.** 45

What number goes in the box?

$$20 \div \square = 4$$

$$\square = \underline{\hspace{10em}}$$

What number goes in the box?

$$\square \times 3 = 27$$

$$\square = \underline{\hspace{10cm}}$$

SET 42 • 6.EE.A.2.a

BACKWARD 3

Order matters in subtraction and division • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$7 - 3 = 3 - 7$$

A. True

B. False

True or false?

$$6 \times 4 = 4 \times 6$$

A. True

B. False

True or false?

$$12 \div 3 = 3 \div 12$$

A. True

B. False

Which statement is true?

A. $10 \div 2 = 2 \div 10$

B. $5 + 9 = 9 + 5$

C. $8 - 2 = 2 - 8$

D. $9 - 5 = 5 - 9$

True or false?

$$3 + 11 = 11 + 3$$

A. True

B. False

SET 42 • 6.EE.A.2.a

FORWARD 1

Write and solve equations of the form $px + q = r$ • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A rectangle has a perimeter of 46 inches. Its width is 8 inches.

Write and solve an equation to find its length.

Length: _____

Jo has \$40. She buys 3 shirts that cost the same amount and has \$4 left.

How much does each shirt cost?

Cost: _____

Which equation represents "5 more than 3 times a number is 26"?

A. $3(n + 5) = 26$

B. $5n + 3 = 26$

C. $3n + 5 = 26$

D. $3n - 5 = 26$

Twice a number minus 7 is 15.

Write and solve an equation to find the number.

Number: _____

A phone plan costs \$30 per month plus \$0.10 per text.
One month's bill is \$42.

How many texts were sent?

Texts: _____

SET 42 • 6.EE.A.2.a

FORWARD 2

Construct a function from a description • Grade 8 • Functions • 8.F.B.4

A pool has 500 gallons of water. It is filled at 25 gallons per minute.

Write a function for the gallons G after t minutes.

$G =$ _____

Tickets cost \$4 each plus a \$6 fee for the order.

Which function gives the total cost y for x tickets?

A. $y = 10x$

B. $y = 4x + 6$

C. $y = 6x + 4$

D. $y = 4x - 6$

The table shows a linear function.

Write an equation for the function.

x	0	1	2
y	7	10	13

$y =$ _____

A line passes through $(0, -2)$ and $(3, 4)$.

Write an equation for the line.

$y =$ _____

True or false?

The function $y = 15x + 20$ has an initial value of 15.

- A. True
- B. False

Identify parts of an expression: term, sum, product, factor, quotient, coefficient

MAIN	Grade 6 main question	6.EE.A.2.b
BACKWARD 1	Sum, difference, product, and quotient	5.OA.A.2
BACKWARD 2	Factors	4.OA.B.4
BACKWARD 3	Interpreting expressions without evaluating	5.OA.A.2
FORWARD 1	Combine like terms with rational coefficients	7.EE.A.1
FORWARD 2	Interpret the parts of a linear function	8.F.B.4

SET 43 • 6.EE.A.2.b

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.b

In the expression $4(9 + 3)$, how can the 4 be described?

- A.** A sum
- B.** A factor
- C.** A term of $9 + 3$
- D.** A quotient

How many terms are in the expression $3x + 5y - 2$?

Terms: _____

What is the coefficient of y in $4x + 9y$?

Coefficient: _____

True or false?

In $5(m + n)$, the quantity $(m + n)$ is a single factor and also a sum of two terms.

A. True

B. False

In the expression $7m + 3$, which part is the constant term?

- A. 3
- B. $7m$
- C. m
- D. 7

SET 43 • 6.EE.A.2.b

BACKWARD 1

Sum, difference, product, and quotient • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

True or false?

$8 + 7$ is a sum.

A. True

B. False

What is 6×9 called?

- A.** A quotient
- B.** A product
- C.** A sum
- D.** A difference

Is $24 \div 6$ a sum, difference, product, or quotient?

Answer: _____

True or false?

15 – 4 is called a product.

A. True

B. False

Which expression is a quotient?

A. 20×5

B. $20 + 5$

C. $20 - 5$

D. $20 \div 5$

SET 43 • 6.EE.A.2.b

BACKWARD 2

Factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

True or false?

In $4 \times 7 = 28$, the numbers 4 and 7 are factors of 28.

A. True

B. False

Name the two factors in 5×9 .

Factors: _____

Which number is NOT a factor of 20?

- A.** 4
- B.** 5
- C.** 6
- D.** 10

True or false?

3 is a factor of 12.

A. True

B. False

Write one factor pair of 18.

Factor pair: _____

SET 43 • 6.EE.A.2.b

BACKWARD 3

Interpreting expressions without evaluating • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

True or false?

$4 \times (2,365 + 718)$ is four times as large as $2,365 + 718$.

A. True

B. False

How does $5 \times (26 + 4)$ compare to $26 + 4$?

- A.** It is the same.
- B.** It is 5 times as large.
- C.** It is 5 more.
- D.** It is 5 less.